American Economic Review 2025, 115(5): 1485–1519
https://doi.org/10.1257/aer.20230524

Efficiency in Household Decision-Making: Evidence from the Retirement Savings of US Couples†

By Taha Choukhmane, Lucas Goodman, and Cormac O'Dea*

We study how couples allocate retirement-saving contributions across each spouse's account. In a new dataset covering over a million US individuals, we find retirement contributions are not allocated to the account with the highest employer match rate. This lack of coordination—which goes against the assumptions of most models of household decision-making—is common, costly, persistent over time, and cannot be explained by inertia, auto-enrollment, or simple heuristics. Complementing the administrative evidence with an online survey, we find that inefficient allocations reflect both financial mistakes as well as deliberate choices, especially when trust and commitment inside the households are weak. (*JEL* D13, G51, J26, J32)

Most households are formed of multiple people. To study their decisions, economists must take a stance on how members of the same household resolve their conflicting desires. Whether households' decision-making results in efficient outcomes—where no resources are wasted in any given period—is not just a core assumption of many influential models of intrahousehold decision-making; it also affects their ability to take full advantage of the economic opportunities available to them.

This paper offers a transparent nonparametric test of the efficiency of married couples' decisions in a high-stakes setting: saving for retirement. Nearly two-thirds of US workers have access to an employer-sponsored defined contribution (DC)

*Choukhmane: MIT Sloan School of Management (email: tahac@mit.edu); Goodman: US Department of the Treasury (email: lucas.goodman@treasury.gov); O'Dea: Yale University (email: cormac.odea@yale.edu). Stefano DellaVigna was the coeditor for this article. We are grateful to the coeditor, Joe Altonji, Orazio Attanasio, John Bailey Jones, James Choi, Andrew Hertzberg, Tatiana Homonoff, Shelly Lundberg, Costas Meghir, Pierre-Carl Michaud, Paolina Medina, Alberto Rossi, Arthur Seibold, Dmitry Taubinsky, and Alessandra Voena for very helpful discussions and to Jun-Davinci Choi, Jackson Howell, Rosa Kleinman, Elijah Olson, Aneesha Parvathaneni, and Alexander-Minh Vu for excellent research assistance. The research reported herein was performed pursuant to grant RDR18000003 from the US Social Security Administration (SSA) funded as part of the Retirement and Disability Research Consortium. The findings, interpretations, and conclusions expressed are solely those of the authors and do not represent the opinions or policy of SSA, the US Department of the Treasury, any other agency of the federal government, or NBER. Neither the United States government nor any agency thereof, nor any of their employees, makes any warranty, express or implied, or assumes any legal liability or responsibility for the accuracy, completeness, or usefulness of the contents of this report. Reference herein to any specific commercial product, process, or service by trade name, trademark, manufacturer, or otherwise does not necessarily constitute or imply endorsement, recommendation, or favoring by the United States government or any agency thereof. We are grateful to the Yale Economics Tobin Center for Economic Policy for providing co-funding. This research was conducted while Goodman was an employee at the US Department of the Treasury. Any taxpayer data used in this research were kept in a secured IRS data repository, and all results have been reviewed to ensure that no confidential information is disclosed.

†Go to https://doi.org/10.1257/aer.20230524 to visit the article page for additional materials and author disclosure statement(s).

retirement saving plan (Topoleski and Myers 2020), and over four-fifths of these plans offer an employer "match" (Arnoud et al. 2021). Because match schedules vary across employers, the financial incentives for saving differ across each spouse's account. We ask whether married couples allocate their retirement contributions to efficiently exploit the match incentives available at the household level. For instance, if one spouse has a dollar-for-dollar employer match up to a cap and the other spouse has a 50 cents-on-the-dollar match, then the efficient allocation at the household level is to fully exploit the match offered to the first spouse before making any contribution to the second spouse's account.

To study this question, we create a new dataset of employer-sponsored retirement plan characteristics. This employer dataset is generated by hand-coding narrative descriptions contained in the annual regulatory filings of over 6,000 DC retirement plans, covering approximately 40 million employees. We link this employer dataset to administrative records on employee retirement saving choices. Our employee data come from IRS tax returns filed by individuals (which allow us to link spouses together) and W-2 forms filed by employers (which report yearly contributions by each employee to these plans). In this matched employer-employee dataset, match schedules vary substantially across employers, which creates an ideal laboratory to study the efficiency of households' financial decisions. The incentives created by the employer match are large and easily measurable, and the efficient allocation for a couple that achieves efficiency can be clearly defined. The setting that we study generates favorable conditions for coordination to emerge; for example, retirement assets are considered marital property in the case of a divorce.

We find that close to 20 percent of couples in our sample fail to exploit a within-period intrahousehold arbitrage condition. That is, these couples could increase their retirement wealth without changing their consumption (or increase their consumption at no cost to retirement wealth) by simply reallocating existing contributions from the account of the spouse with a lower marginal match incentive to the account of the spouse with a higher marginal match incentive. The roughly four-fifths of couples who do not fail this efficiency test are not necessarily actively coordinating: They may happen to (independently) choose individual contributions that are consistent with household-level efficiency (e.g., both could be fully exploiting their employer match). To provide a benchmark for the extent of inefficient allocations absent any coordination between spouses, we compare real couples to randomly drawn pairs of individuals with similar observable characteristics. In these placebo samples—in which there is (by construction) no coordination—we find that only 33 percent to 34 percent of placebo couples fail to exploit an available arbitrage opportunity. Relative to these benchmarks, our finding of 19.3 percent of married couples having some forgone match is consistent with some couples actively coordinating their contributions and a majority not doing so.

The mean and median levels of forgone match for couples who fail to exploit the intrahousehold arbitrage opportunity are substantial, at $757 and $383 per year, respectively. In addition, inefficiency is persistent: More than half of couples with an inefficient allocation in a given year still allocate their savings inefficiently four years later. Relative to a counterfactual in which all couples saved the same amount but allocated their contributions efficiently, we estimate in a simple simulation

exercise that noncoordination throughout the life cycle can lower wealth at retirement by an average of approximately $14,000.

Next, we explore mechanisms and evaluate the extent to which the lack of coordination we document is driven by deliberate noncoordination and the extent to which it is due to mistakes. We complement our administrative data with a survey in which married people were asked to make a hypothetical allocation between their account and their spouse's account, which has a different match rate. For the 40 percent of survey respondents who do not maximize the overall match their couple could have achieved, we follow up with a set of questions designed to evaluate whether the decision was a deliberate one (i.e., respondents *knowingly* left money on the table) or an accidental one (indicated by respondents wishing to reallocate their contributions to maximize the match once we alert them to the implications of their choice). We find that approximately half of the incidence of forgone match in our survey can be characterized as deliberate, and half as accidental.

For drivers of accidental forgone match, using our administrative data, we find no evidence that rational inattention, inertia, or simple savings allocation heuristics explain the noncoordination that we document. Couples do not systematically improve efficiency when they make active savings decisions—evidence against inertia driving our results. We do find evidence of the use of a savings heuristic whereby couples equalize their contributions, but this strategy acts to facilitate efficiency rather than driving inefficiency. We also show that noncoordination is insensitive to the stakes of coordination and that it persists even when there is more than $6,000, or 5 percent of joint earnings, at stake. We interpret this finding as evidence that our results are not driven by rational inattention. However, using our survey, we show that inattention (albeit perhaps not of a rational type) is a contributing factor: The vast majority of those who had hypothetical forgone match without realizing it reallocated their contributions once we alerted them to the implications of their choice. Additionally, among the respondents to our survey for whom both they and their spouse have DC plans in real life, over one-third stated that they had not considered that there might be gains to coordination.

Turning to drivers of deliberate forgone match, we note that Chiappori and Mazzocco (2017) suggest that efficient coordination can be facilitated by the "proximity and durability of the relation." Consistent with this mechanism and using our administrative data, we find that plausible proxies for marital commitment are correlated with efficient coordination. Conditional on a couple's observable characteristics, the incidence of forgone match is higher for couples who our data show will subsequently divorce and is lower for couples whom we observe to have a joint bank account in the years before marriage. Given that wealth in retirement accounts is a marital asset, this could be seen as a puzzle: Even if divorce is *certain*, there are incentives to avail of the arbitrage opportunity. Our survey reveals, however, that close to one-fifth of couples say that they do not know how their retirement wealth would be divided upon divorce, and over one-third believe that they would keep their own accounts. Those who believe the latter are also substantially more likely to have forgone match in the hypothetical choice problem than those who believe retirement wealth is split on divorce.

We make contributions to two branches of the literature. First, our paper is related to a large theoretical and empirical literature on intrahousehold decision-making.

On the theoretical side, both the unitary model and the broader class of collective household models make the minimally restrictive assumption that decisions made in each time period are Pareto efficient—that is, no available resources are wasted within the household (see Chiappori and Meghir 2015 and Chiappori and Mazzocco 2017 for reviews).[1] Our results of widespread inefficiency are suggestive of a greater role for intertemporal models of the household that do not assume that couples achieve ex post efficiency in every period. For example, Basu (2006) and Hertzberg (2024) develop models in which equilibrium behavior leads to inefficient outcomes. Models have also been proposed in which there exists a noncooperative threat point in the bargaining problem (see, for example, Lundberg and Pollak 1993; Del Boca and Flinn 2012; and Browning, Chiappori, and Lechene 2010). In such models, couples may behave inefficiently when the threat point is realized.

On the empirical side, a large number of studies have tested the efficiency of decisions using household data from many developed and developing countries (Bourguignon et al. 1993; Chiappori, Fortin, and Lacroix 2002; Blundell et al. 2007; Cherchye, De Rock, and Vermeulen 2009; Dauphin et al. 2011; Attanasio and Lechene 2014; LaFave and Thomas 2017; and Bargain et al. 2022). While results and approaches differ between papers, most of these studies have failed to reject the efficiency of married couples' decisions using survey data of consumption. Dauphin, Fortin, and Lacroix (2018) suggests that these consumption tests in small samples may be underpowered to reject household efficiency.

A number of papers have tested the efficiency of household investment decisions in developing countries and found more mixed results in both observational and field-experimental data. Udry (1996) finds that agricultural plots controlled by women in Burkina Faso are farmed less intensively than plots controlled by their husbands and that a reallocation of inputs could be Pareto improving.[2] However, other interpretations consistent with household efficiency have been offered: Differences in investment intensity could reflect gender differences in the security of land rights (Goldstein and Udry 2008) or measurement error (Rangel and Thomas 2019). These empirical challenges can be overcome in field experiments. While experimental evidence supporting efficiency exists (e.g., Bobonis 2009), there is also ample experimental evidence of inefficient outcomes. Ashraf (2009); Schaner (2015); and Almås et al. (2018) all find evidence of spouses preferring lower levels of resources that they control over higher levels of resource with shared control.[3] These field-experimental results provide powerful evidence that inefficiencies can occur, but the extent to which these findings extend to naturally occurring high-stakes financial decisions in which cooperation could emerge after repeated interactions remains an open question.

Our contribution to that literature is to implement a transparent nonparametric test of household-level efficiency in a naturally occurring setting, in which the

[1] Dynamic implementations of the collective model (e.g., Ligon 2002; Mazzocco 2007) confront the fact that individuals cannot commit to future behavior. These models admit outcomes that are inside the Pareto frontier that would be attainable if couples could fully commit to future actions. But they still retain the property that in every period no surplus is left on the table. It is this (weaker) notion of ex post, or static, efficiency that we test.

[2] See Walther (2018) for similar recent evidence from Malawi.

[3] There is additional experimental evidence of inefficient household decision-making in settings other than saving choices. For example, see recent work on information aggregation in India (Conlon et al. 2021).

incentives are directly measurable to the researcher and relatively simple to the individuals (i.e., unlike with agricultural investments, the rate of return is given by a straightforward deterministic formula). The substantial level of inefficiency that we document is all the more notable given that our setting creates favorable conditions for cooperation to emerge: (i) our sample of married tax filers with access to two employer-sponsored DC accounts earns approximately twice as much as the average US household and, as such, is presumably more educated and financially literate; (ii) retirement saving is a repeated decision, and couples have time to learn and build familiarity with the setting; and (iii) retirement assets are relatively illiquid before retirement and, across all US states, are divided independently of who made the contribution in the case of a divorce, which should alleviate commitment frictions.

We also contribute to a growing literature on suboptimal financial decision-making by households. This includes evidence that households do not locate their financial assets efficiently across their taxable and nontaxable savings accounts (Bergstresser and Poterba 2004), fail to refinance a fixed-rate mortgage when it is beneficial to do so (Andersen et al. 2020), and co-hold low-interest liquid savings and high–interest rate credit card debt (Gross and Souleles 2002). While some of these apparent mistakes could be rationalized depending on household circumstances, evidence of unambiguous financial mistakes (e.g., no-arbitrage violations) is more limited. Leading findings include older workers not taking advantage of a 401(k) match despite being eligible for immediate penalty-free withdrawals (Choi, Laibson, and Madrian 2011) and credit card borrowers not prioritizing the repayment of the balance with the highest interest rate (Gathergood et al. 2019).[4] We add to this literature by documenting a new instance of inefficient financial behavior with a large and persistent cost. The average annual cost of inefficient behavior in our setting is an order of magnitude larger than in the credit card setting studied by Gathergood et al. (2019) and affects a larger share of the population than the setting studied by Choi, Laibson, and Madrian (2011) (which is limited to workers older than 59.5 in plans that do not restrict in-service withdrawals). We show a relationship between the noncoordination we document and financial decision-making and so contribute to a burgeoning literature that has linked poor financial literacy to suboptimal decision-making (see Lusardi and Mitchell 2023 for a recent review).

The paper proceeds as follows. Section I formalizes our test of efficiency and motivates our empirical approach. Section II describes our data. Section III contains our results on the incidence of noncoordination. Section IV investigates the drivers of these patterns. Section V concludes.

I. Framework and Empirical Approach

In this section, we motivate and formalize our nonparametric test of Pareto efficiency.

[4] Goodman, Mukherjee, and Ramnath (2023) find another example of an unambiguous mistake in retirement saving: 0.4 percent of retirement-age savers forgot about or abandoned their IRA accounts.

A. *Framework*

We consider a couple with members $i \in \{A,B\}$, who live for two periods $t \in \{1,2\}$. Each member of the couple earns an income in the first period y_1^i and no income in the second period. Saving done by individual i in period 1 (s^i) yields a level of wealth available for consumption in period 2 of $R \times \left[s^i + m^i\left(s^i\right)\right]$, where $R \times \left[s^i + m^i\left(s^i\right)\right]$ is the technology that converts saving to wealth. This comprises a gross investment return R, assumed identical between the members of the household, and the employer match $m^i\left(s^i\right)$, which may differ between the two spouses and may be nonlinear.

The household chooses consumption and saving for each individual in each period, subject to an intertemporal budget constraint:

$$\sum_{i=A,B}\left(c_1^i + c_2^i\right) \leq \sum_{i=A,B}\left(\left(y_1^i - s^i\right) + R \times \left[s^i + m^i\left(s^i\right)\right]\right). \tag{1}$$

Household preferences could be characterized using a single utility function (as in a unitary model) or as a weighted sum of individual utility functions (as in a collective model of Chiappori 1988 and Browning and Chiappori 1998). These are the two workhorse models that have been used for empirically studying household decision-making.[5] In either case, however, as long as these utility functions are always increasing in consumption, and the framework is one that assumes "ex post" (or "static") efficiency[6], the aggregate saving in the first period must be done in a fashion that maximizes the employer match. That is, letting S be the total amount of saving ($s^A + s^B$), the optimal allocation of saving across spouses $\left\{s^{*A}(S), s^{*B}(S)\right\}$ must satisfy

$$\left\{s^{*A}(S), s^{*B}(S)\right\} \in \arg\max\, m^A\left(s^A\right) + m^B\left(s^B\right) \tag{2}$$

subject to

$$s^A + s^B \leq S.$$

We can define the excess of maximum possible match over the actual match as the "forgone match" (FM):

$$FM = \left[m^A\left(s^{*A}(S)\right) + m^B\left(s^{*B}(S)\right)\right] - \left[m^A\left(s^A\right) + m^B\left(s^B\right)\right]. \tag{3}$$

[5] The unitary model implies that proxies of spouses' bargaining power do not affect household choices. Evidence exists that they do (Schultz 1990; Thomas 1990; Lundberg, Pollak, and Wales 1997; Aura 2005; Addoum 2017; Cesarini et al. 2017). The collective model can accommodate this in a framework that assumes efficiency. Recent papers in the collective spirit include the study of the impact of divorce laws (Voena 2015; Reynoso 2024); educational choices by gender (Bronson 2014); the interplay between marriage, education, and labor supply (Chiappori, Dias, and Meghir 2018); welfare reform (Low et al. 2018); portfolio choice (Gu, Peng, and Zhang forthcoming); fertility decisions (Low 2024; Doepke and Kindermann 2019); intrahousehold allocation (Lise and Yamada 2019); intertemporal discounting (Adams et al. 2014); joint leisure choice (Cosaert, Theloudis, and Verheyden 2022); and sorting in the marriage market (Calvo, Lindenlaub, and Reynoso 2024).

[6] Limited commitment models (e.g., Mazzocco 2007), which assume individuals cannot commit to future behavior, admit outcomes that are ex ante inefficient (that is, they are inside the Pareto frontier, which exists at the time the couple forms) but retain the assumption that, in each time period, no surplus is left on the table.

The condition $FM = 0$ is testable given variation within households in the marginal savings technology of individuals, equal to $R\frac{dm^i}{ds^i}$. Our setting provides such variation, both because spouses may face different match schedules m and because these match schedules are, in general, nonlinear.

B. *Empirical Approach*

Our test of household efficiency requires us to measure the amount of forgone match for each household in a sample. The empirical requirements for such an exercise are a dataset that (i) links individuals in married couples; (ii) contains details of saving at the *individual* level s^A, s^B; and (iii) measures match schedules, also at the *individual* level $(m^A(\cdot), m^B(\cdot))$. With such a dataset, which we create, the calculation of *FM* using equations (2) and (3) is straightforward. We report results using these calculations in Section III. Finding $FM > 0$ for a particular couple indicates that a couple is not allocating contributions in an efficient fashion.

C. *The Role of Divorce and Death*

While the framework outlined does not incorporate marital separation (in the form of divorce or death), the legal system offers strong protections for spouses after such events. These protections create strong incentives for spouses to coordinate their contributions even when facing the threat of separation. Taking death first, under the Retirement Equity Act (1984), a spouse must be the beneficiary of the DC plan, unless they provided written consent to waive their entitlement. As regards divorce, across all US states, the disposition of retirement accounts, in the absence of a prenuptial agreement, is not influenced by which spouse made contributions. There is no direct influence because it is generally assumed that married parties cooperate as a partnership in determining which roles each spouse should take on to maximize their income, and thus, each spouse should share equitably in the accumulated assets. This implies that couples have an incentive to maximize their joint retirement wealth even if they are certain to divorce later on.

D. *Incentives to Prefer Contributions to Own Account*

Even if the division of retirement wealth on divorce in the United States is generally unaffected by who made the contribution, there could remain strategic reasons for a spouse to prefer to contribute to their own account rather than their spouse's account with a higher match. First, a prenuptial agreement might have been signed, carving out retirement wealth as personal property (but, arguably, such agreements could be revised after the marriage—with a postnuptial agreement—to ensure the couple can take advantage of available arbitrage opportunities). Second, understanding of divorce law might be imperfect (and in Section IVC, we provide survey evidence that it is). Third, having funds in one's own account might bring a greater influence over the use of the funds in retirement or more power in the marriage more generally. Such strategic considerations could rationalize a spouse saving at a match that is lower than the potential match available to their spouse.

Strategic considerations by themselves do not mean that two spouses could not profitably benefit from this arbitrage opportunity. Consider spouse A, with a match rate of m^A in period 1, who wishes to have a saved amount $\left(1 + m^A\right)s$ available in period 2; she can achieve this by saving s in her account. If spouse B has a higher (unused) match rate of m^B, then spouse A could generate a surplus $\left(m^B - m^A\right)s$ by contributing s to spouse B's account instead of her own. This surplus can be divided between the two spouses, leaving both of them strictly better-off in period 1. For this trade to be Pareto optimal, spouse B must then transfer back the savings $\left(1 + m^A\right)s$. Enforcing these transfers in period 2 could be done using a postnuptial or other contractual arrangement or—within a repeated game—through a punishment strategy of spouse A that enforces cooperation. In some relationships, marital commitment will be sufficient to enforce coordination, in others perhaps not, and below, we document positive correlations between plausible proxies for commitment and cooperation.

II. Data

Our administrative data bring together newly constructed employer data on retirement plan characteristics—matching schedules, vesting schedules, and "auto-features"—and employee data on retirement saving. The next two subsections discuss each in turn.

A. *Retirement Plan Data*

The Employee Retirement Income Security Act (1974) is a federal law that governs the provision of employee benefits, including retirement plans. Compliance with the act requires an annual report from all firms with retirement plans. The reporting involves submitting a Form 5500, which reports, for example, the type of plan offered (defined benefit or defined contribution), the total number of participants, aggregate employer contributions, and aggregate employee contributions. Match schedules and vesting schedules are not collected as part of the regulatory form, but plans with more than 100 participants are also obliged to submit an auditor's report that contains a narrative "description of the plan," which must describe in free form text, among many other details, the plan's match schedules (if any), its vesting schedules (if any), and its auto-features (if any). These narrative retirement plan descriptions are publicly available to download.[7]

To create a new dataset, we extract and hand-code matching schedules, vesting rules, and auto-features from these narrative descriptions. Full details are given in Supplemental Appendix A.1; briefly, the approach involves first finding the relevant key passages by identifying key words and phrases (e.g., "description of the plan," "matching," "vesting," etc.) and then extracting the relevant pages for a sample of firms before finally reading the retirement plans and codifying them into a dataset.

The retirement plan data used in this paper are formed by codifying the plan characteristics of over 6,000 401(k) and 403(b) plans in the United States. The bulk of

[7] www.efast.dol.gov/5500search/

this sample is comprised of the largest approximately 5,000 plans, where we define size of plan as the mean number of participants over the period from 2003 to 2018. We also codified the details of a random sample of smaller plans. Our data are longitudinal; we have hand-collected retirement plan characteristics for each year over that period, yielding over 70,000 plan year observations.[8] A concern with any hand-collected data is that the data could be contaminated by measurement error. In Supplemental Appendix B, we provide several pieces of evidence that our results are not driven by measurement error.

The three key retirement plan features on which we collect data are match schedules, vesting schedules, and auto-features. We also collect data on whether a single schedule of plan details applies to all members or whether different plan features are offered to different categories of worker: Our linking of employee to plan requires that all employees have access to the same plan, and we define our analysis sample accordingly. Matching schedules, the piece of retirement plan data at the heart of our test of efficiency, are piecewise linear functions that determine the contribution employers make to their employees' accounts. Supplemental Appendix Figures A1 and A2 illustrate these match schedules.

B. *Combining with Employee Savings Data*

Our data on earnings and retirement plan contributions come from tax return data (Internal Revenue Service 2024). In particular, Form W-2 identifies the DC contributions made by a given employee at a particular job (identified by the EIN on the W-2). We develop a crosswalk to map from EINs reported on Form 5500 to EINs reported on Form W-2; Supplemental Appendix A.2 gives further details.

For our empirical approach, it is very important to correctly match a given worker to a given DC plan, so we drop firms that substantively offer more than one plan, e.g., one plan for a certain class of employees and another plan for a different class.

C. *Defining Our Population*

The population we study is married couples in the United States who satisfy four restrictions. First, they must file a tax return jointly. Second, both spouses must have positive wages. Third, both spouses must be employed at firms that offer an employer-sponsored defined contribution plan. Fourth, both spouses must be at least 21 years of age.[9] These restrictions leave us with a study population that contains approximately one-third of the entire population of married US tax filers. This study population differs systematically from the broader US population. Panel A of Table 1 shows, for 2015, mean and median income, mean age, and mean duration (up to that year) of the marriage. It shows median income for our study population is approximately $101,000. This is close to twice the median household income of

[8] For more on this process and our new dataset, see Arnoud et al. (2021), which uses a subset of the data. This data is available in the repository for Choukhmane et al. (2025). Other papers that codify details in the Form 5500 narratives include Bubb and Warren (2020) and Rauh, Stefanescu, and Zeldes (2020).

[9] While firms are disallowed by ERISA law in restricting eligibility to certain classes of employees, they are allowed to exclude those aged under 21 from participating.

Table 1—Summary Statistics on Employee Data

	Income ('000s)					Share federal	
	Mean	Median	Age	Marriage length	Pop. size	Either	Both
Panel A. Population							
	$136.1	$101.3	43.8	11.1	27,381,700	0.046	0.005
Panel B. Matched sample							
	$147.3	$119.3	44.4	10.9	504,831	0.435	0.284
Panel C. Analysis sample							
1. Analysis sample: Full	$159.2	$129.8	45.1	11.4	424,278	0.487	0.323
2. Analysis sample: Baseline	$175.7	$143.6	45.1	11.9	184,619	0.578	0.448

Notes: Panel A shows summary statistics on the couples in the population we define for 2015, who are those couples in the United States who satisfy four requirements: (i) they file a tax return, (ii) both spouses are employed, (iii) both spouses have access to a DC plan, and (iv) both spouses are at least 21 years of age. Panel B gives summary statistics on our sample in 2015. These are those couples where both spouses are members of retirement plans in our plan dataset. Panel C shows summary statistics for our analysis sample. Row 1 gives our "full" analysis sample, which restricts to couples where at least one spouse makes a DC contribution and at least one spouse works for an employer that offers a match—as these are the couples for which our empirical test can be performed. Row 2 restricts to our "baseline" sample in 2015, which is a subset of our full sample where both members of the couple are vested in their retirement plan, have at least two years of tenure, and do not have a substantial age gap (details of this final restriction are discussed in the text). The rationale for these restrictions is discussed in Section III. "Income" is adjusted gross income. "Marriage length" is censored above at 19 due to data limitations. To protect taxpayer privacy, medians are calculated as pseudomedians, equal to the mean of the 20 observations nearest the true median.

the population that year. The population that we study is on average, therefore, substantially better-off than couples in the US population.

Our test of efficiency requires both employee data on savings and employer data on plan details. To maximize the size of our analysis sample, as discussed in the previous subsection, we chose to code the retirement plan data of the largest private and nonprofit sector plans. We also code the plan of the federal government, though we exclude all federal employees whose W-2 comes from military payers (as uniformed members of the military were not eligible for the match in 2015). Our sample is therefore large, representing approximately 40 million individual employees, a substantial proportion of the US workforce and an even more substantial share of those in the US workforce who have access to an employer-sponsored defined contribution plan. Our matched sample is comprised of couples where both spouses (i) are in the population we defined above, (ii) are in our individual merged dataset, and (iii) are members of plans where there is a single match schedule (ensuring we know the match schedule that pertains to them). This yields a sample of approximately 505,000 couples. Panel B of Table 1 shows summary statistics for the couples in this matched sample. Differences between our merged sample and the underlying population we have set out to study are modest.

To form a sample of couples on whom our test can be carried out, we make two further restrictions. We require that a least one spouse makes a DC contribution and at least one spouse works for an employer that offers a match. Applying these restrictions to our matched sample gives our full "analysis" sample, summary statistics for which are given in the first row of panel C. Row 2 further restricts the sample to a "baseline" sample, which makes three further restrictions designed to eliminate couples where our empirical test is somewhat less clean. First, we remove couples where either spouse has insufficient tenure to be fully vested in their employer's

contributions; this ensures that forgone match is not driven by differing probabilities of forfeiture between spouses. We can do this as we have codified the vesting rules in our plan data and can measure tenure in the administrative data. Second, we remove spouses where, due to an age difference, there is a difference across spouses in how liquid a plan is. In general, withdrawals before the age of 59.5 are subject to a 10 percent tax penalty. For this reason, we drop couples where one spouse is older than 59.5 and the other is younger; we also drop couples where both spouses are younger than 59.5 but one spouse is considerably older than the other.[10] Such a difference in age could make contributing to an account with a lower marginal match rate favorable if it is likely to become liquid much sooner. Third, we drop couples where either member had tenure of one year or less. We do this as firms are allowed to exclude employees who have less than one year's tenure from their retirement plan.[11]

Unless otherwise stated, our analysis uses a single cross section (2015). In some analyses, we exploit the panel dimension of our data, and we give details of this where relevant below.

III. Results

In this section, we present the results from our test of efficiency. We show that roughly one in five couples allocates their savings inefficiently and that such noncoordination is persistent over time, costly, and pervasive across subsamples defined by the characteristics of the plan and of couples. In order to interpret the magnitude of these results, in Section IIIC we form benchmarks for how much inefficiency we should expect absent any coordination between spouses. A comparison of our results with these benchmarks suggests noncoordination of financial decision-making is widespread.

A. *Noncoordination Is Common, Persistent, and Costly*

A Motivating Example.—We start with a particular subsample of our data where the efficient strategy is simple: One spouse should do *all* the saving. We impose two conditions to guarantee that this is the unique match-maximizing strategy. First, each spouse must face a different match rate on their first dollar of saving. We refer to the spouse with lower and higher match rates as, respectively, L and H. The second condition is that the couple's total saving must be weakly less than the first match cap of spouse H (that is, the point at which the match rate falls, either to zero or to a lower match rate).[12]

Figure 1 summarizes the extent of inefficiency for the sample who satisfy these two conditions. The figure uses two metrics: the share of saving done by the spouse

[10] Specifically, we drop couples where (59.5 minus the age of the younger spouse) is more than twice (59.5 minus the age of the older spouse).

[11] While these three restrictions are a priori important, they actually make little difference to our headline result.

[12] Out of 184,600 couples in our baseline sample, these restrictions exclude all but 2,800. Beginning in Section IIIA, we expand our analysis to the full set of couples.

Panel A. Prediction: Savings share of high-match spouse

Match rate of low-match rate spouse (percent) \ Match rate of high-match rate spouse (percent)	0	(0, 50)	50	(50–100)	[100, max]
0	?	100	100	100	100
(0, 50)		?	100	100	100
50			?	100	100
(50–100)				?	100
[100, max]					?

Panel B. Observed: Savings share of high-match spouse

Match rate of low-match rate spouse (percent) \ Match rate of high-match rate spouse (percent)	0	(0, 50)	50	(50–100)	[100, max]
0	–	86	81	84	76
(0, 50)		–	71	77	67
50			–	70	55
(50–100)				–	63
[100, max]					–

Panel C. Prediction: Share of couples with forgone match

Match rate of low-match rate spouse (percent) \ Match rate of high-match rate spouse (percent)	0	(0, 50)	50	(50–100)	[100, max]
0	?	0	0	0	0
(0, 50)		?	0	0	0
50			?	0	0
(50–100)				?	0
[100, max]					?

Panel D. Observed: Share of couples with forgone match

Match rate of low-match rate spouse (percent) \ Match rate of high-match rate spouse (percent)	0	(0, 50)	50	(50–100)	[100, max]
0	–	24	30	31	31
(0, 50)		–	47	48	44
50			–	59	57
(50–100)				–	49
[100, max]					–

Panel E. Average forgone match

Match rate of low-match rate spouse (percent) \ Match rate of high-match rate spouse (percent)	0	(0, 50)	50	(50–100)	[100, max]
0	–	197	532	807	1,014
(0, 50)		–	268	814	716
50			–	298	570
(50–100)				–	435
[100, max]					–

FIGURE 1. PATTERNS IN SIMPLEST CASE

Notes: This figure restricts to the subset of couples where the only efficient allocation is to save entirely in one spouse's account; this spouse is denoted as spouse *H*, and the other spouse *L*. This restriction leaves only a small share of our sample: 2,800 observations. Panels A and B compare the theoretical predictions and empirical findings for the share of saving allocated to spouse *H*'s account. Panels C and D compare the theoretical predictions and empirical findings for the presence of forgone match. Panel E reports the average forgone match (conditional on positive forgone match). Forgone match is defined as the difference between (i) the match that would be received if total contributions were allocated to maximize the total match received and (ii) the match actually received.

with the higher match rate and the proportion of couples with some forgone match.[13] Efficiency would imply that these would be 100 percent and 0 percent, respectively, illustrated in panels A and C of Figure 1. In all exhibits, the match rate of spouse *L* is given in the rows, and the match rate of spouse *H* is given in the columns. We group match rates into five groups. The first group has no match, and subsequent groups have a match rate of greater than 0 but less than 50 percent, a 50 percent match, a match greater than 50 percent but less than 100 percent, and a match greater than or

[13] In defining positive forgone match, we apply a de minimis threshold of $10 per year in all our analyses. See Supplemental Appendix A.3 for additional details on how we calculate the forgone match.

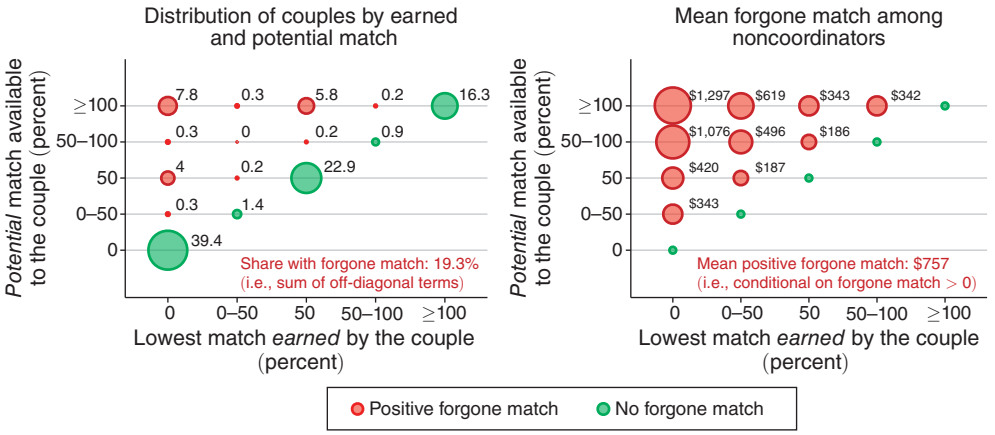


FIGURE 2. THE DISTRIBUTION OF COUPLES BY EARNED AND POTENTIAL MATCH

Notes: We define the "lowest match earned" by the couple as the lowest match rate earned by either spouse on their last dollar of contribution. We define the "potential match available to the couple" as the largest match rate that could be earned by either spouse on their next dollar of contribution. We assign couples that have no potential match because they are both constrained by annual contribution limits to the diagonal corresponding to the highest match earned. We similarly assign couples whose forgone match is below the de minimis threshold of $10 to the diagonal corresponding to their highest match earned; 0.2 percent of those couples on the diagonal nevertheless have forgone match (beyond the de minimis threshold) due to the binning of the x- and y-axes in the figure. Forgone match is defined as the difference between (i) the match that would be received if total contributions were allocated to maximize the total match received and (ii) the match actually received.

equal to 100 percent. Panel B of Figure 1 gives the proportions saved by spouse *H* in reality. The proportions range from 55 percent to 86 percent, meaningfully different from the theoretical benchmark of 100 percent. Panel D of Figure 1 shows that, in reality, the proportion of couples with a forgone match ranges from 24 percent to 59 percent, far from the efficient outcome of 0 percent.

Noncoordination Is Common.—The analysis in the motivating example above uses a small and select group of couples. We excluded couples with the same match schedule; such couples can behave efficiently or can have forgone match if one spouse saves beyond their match cap, while the other does not fully exploit their match. We also only included spouses with saving low enough for the efficient strategy to be for one spouse to do all the saving. Moving to our baseline analysis sample, however, we still find substantial forgone match: 19.3 percent (with a standard error of 0.09 percent) of couples have some forgone match and are, therefore, not coordinating their retirement savings decisions. These couples could receive, on average, an additional $757 a year in employer matching contributions by moving some of their existing savings from the account of the spouse with a lower marginal match rate to the account of the spouse with the higher marginal match rate.

Figure 2 illustrates the combination of saving decisions that lead to this outcome. The left panel plots the distribution of couples by the combination of the (i) smallest match rate that either spouse earned on their last dollar (their *earned match*) and (ii) the highest potential match that either spouse could have earned on their next dollar (*potential match*). For couples represented by a point on the diagonal, whose *earned*

match is equal to the maximum *potential match*, there is no reallocation that would increase their collective match.[14] The most common scenario for efficient allocations, accounting for 39 percent of our couples, occurs when both the last earned match and the potential match are 0. Most such cases are where both spouses are saving more than their match cap. In this situation, the highest potential marginal match available is 0 percent for both spouses, and the allocation is efficient. The most frequent case of noncoordinated allocations, representing 7.8 percent of all couples in our sample, arises when one spouse contributes above their match cap (resulting in a marginal match rate of 0 percent) while the other spouse has access to a marginal match equal to or greater than 100 percent. The proportion of couples with forgone match (19.3 percent) is certainly a lower bound on the proportion of couples not coordinating their financial decisions. Many couples will have no forgone match even if they are not coordinating: As one example of this, both spouses could be independently fully exploiting their employer match. We return to this in Section IIIC.

In cases where a couple earns less than the maximum match available, they miss out on some potential employer-matching contributions. The right panel of Figure 2 shows the average forgone match for couples at each point in the graph. The most frequent case of noncoordinated allocations, when one spouse contributes above their match cap (resulting in a marginal match rate of 0 percent) while the other spouse has access to a marginal match equal to or greater than 100 percent, is also the most costly form of noncoordination, resulting in an average forgone match of $1,297 a year. The second most frequent form of inefficient allocations, accounting for 5.8 percent of couples, arises when one spouse contributes and earns a marginal match rate of 50 percent, when additional contributions to their spouse's account could have earned a match equal to or greater than 100 percent. Because this form of inefficiency induces a smaller wedge between earned and potential matches, it is, on average, less costly and results in a forgone match of $343 a year.

Noncoordination Is Persistent.—We exploit the panel nature of our data to analyze the persistence in having forgone match. We take all observations with positive forgone match from 2003 through 2011 and restrict attention to those that we observe continuously for the subsequent seven years at the same employer. Figure 3 shows that, conditional on having some forgone match in a given year, 70 percent have some forgone match one year later, 61 percent still have some two years later, while 42 percent have some forgone match seven years later.

There are several ways that one can move from having forgone match to not having it. It could be that the couple reallocate their contributions holding saving constant (perhaps as a deliberate step in response to the fact that they had an unexploited arbitrage condition). Only 11 percent of transitions out of forgone match are of this type. Twenty-five percent of transitions are associated with a decrease in the average saving rate, while most (64 percent) occur with an increase in the average

[14] In computing the earned match and potential match, we take into account corners: Individuals cannot reduce their contributions if they are contributing zero, and they cannot increase their contributions if they are contributing the maximum annual limit. If both spouses are contributing at their annual limit, then they have no forgone match; they are assigned to the diagonal associated with the maximum match rate of the couple on their last dollar.

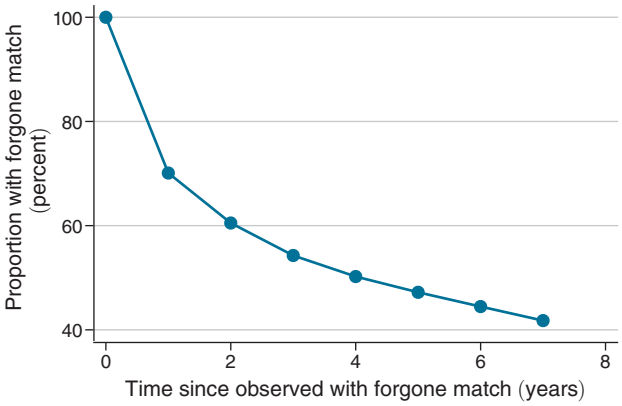


FIGURE 3. PERSISTENCE OF HAVING A FORGONE MATCH

Notes: The sample for this figure is the set of all observations in our panel (subject to our baseline vesting, tenure, and age restrictions) from 2003 to 2011, where (i) there is positive forgone match and (ii) we observe seven consecutive subsequent years of data. We plot the share of observations that experience positive forgone match in years 0 through 7 relative to the initial observation. Forgone match is defined as the difference between (i) the match that would be received if total contributions were allocated to maximize the total match received and (ii) the match actually received.

saving rate.[15] We take from this that the pattern by which couples move out of having forgone match is predominantly not the easiest possible reallocation that they could do to "fix" the forgone match that we document.

Noncoordination Is Costly.—Panel A of Table 2 summarizes the distribution of the annual cost of noncoordination for those couples who exhibit it. While the average forgone match is $757 annually, the underlying distribution is skewed: At the ninetieth percentile, couples fail to capture $1,940 in potential match, whereas the median is $383 annually. These forgone employer matches represent a sizable portion of potential savings for couples, on average equivalent to 13 percent of their household-level employee contributions to retirement accounts, and 8 percent at the median.

To quantify the *lifetime* cost of noncoordination, we build a simple simulation model based on the empirical evidence in our panel sample. We simulate the evolution of forgone match for 100,000 couples between ages 25 and 59 and calibrate the model to match the incidence and the amount of forgone match by age and lagged forgone match status in the preceding 3 years. We assume that the forgone savings would have been invested in a Target Date Fund, the typical investment in employer-sponsored retirement accounts, until age 65.[16] In our simulation, relative to

[15] These statistics discard the 1 percent of such couples where at least one spouse's firm changed their match schedule. We define the average saving rate of the couple as the simple average of the saving rate of the two spouses. We consider the average saving rate to be held fixed when it does not change by more than 0.25 percentage points. In our panel (i.e., without restricting to those escaping noncoordination), 40 percent of couples with positive saving at time $t-1$ hold their savings rate fixed between $t-1$ and t using this 0.25 percentage point threshold.

[16] We assume that savings are invested in and earn the average returns of three asset classes: equities, bonds, and treasury bills. We calibrate the asset allocation by age to match that of Fidelity Target Date Funds. See Supplemental Appendix C for additional details.

Table 2—Distribution of Foregone Match (FM), per Year, for Those Not Coordinating

	Mean	p10	p25	p50	p75	p90
Panel A. Annual cost for couples with positive forgone match						
Dollars per year	$757.2	$58.8	$155.4	$382.6	$886.4	$1,939.5
	($5.7)	($0.9)	($1.6)	($3.2)	($8.0)	($19.8)
As a share of couples' employee contributions	12.6%	1.0%	2.9%	7.8%	17.2%	30.5%
	(0.08%)	(0.02%)	(0.03%)	(0.07%)	(0.09%)	(0.24%)
Panel B. Simulated lifetime cost of noncoordination						
Sum of forgone dollars	$3,778	$0	$0	$1,485	$5,481	$11,243
Forgone wealth at age 65	$13,807	$0	$0	$4,805	$18,917	$41,495

Notes: In panel A, the first row summarizes the distribution of annual forgone match, conditional on forgone match being positive (and subject to a de minimis threshold of $10) in our baseline sample. The second row summarizes the distribution expressed as a proportion of the total employee contributions made by both spouses. For disclosure protection, all percentiles in panel A are "pseudopercentiles," equal to the mean of the 20 observations nearest the true percentile. In panel B, the first row summarizes the distribution of simulated cumulative forgone dollars over an assumed marriage length of 35 years. Row 2 summarizes the distribution of forgone wealth at age 65, under the assumption that forgone dollars would have been invested in a typical retirement investment fund (see Supplemental Appendix C for details). Forgone match is defined as the difference between (i) the match that would be received if total contributions were allocated to maximize the total match received and (ii) the match actually received.

a counterfactual in which all couples saved the same amount but allocated their contributions efficiently, noncoordination results in an average of $13,807 less wealth when retiring at age 65. At the ninetieth percentile, our simulation suggests that noncoordination over an assumed marriage length of 35 years results in $41,495 less wealth when retiring at age 65. These simulation results provide an order of magnitude for the potentially large lifetime negative impact of not coordinating retirement contributions on retirement wealth accumulation.

B. *Noncoordination Occurs across Demographic Groups and Plan Types*

Figure 4 shows how the incidence of forgone match varies along six different characteristics. These are total wages (in quartiles), age (mean couple of age divided into < 35, 35–49, 50+) education (0, 1–4, and 5 or more calendar years of postsecondary education), parental income (in quartiles), the number of federal employees, and the divorce law regime in the couple's state (community property or equitable division).[17] The incidence of forgone match falls modestly with own wages and parental income, and rises modestly with age. Those with the highest levels of education are less likely than those with less to have forgone match, and couples with one federal employee have higher forgone match than those with none or one. Yet forgone match is substantial within all of these cells.

The preceding analysis divides the sample by the characteristics of couple; we also show results in the Supplemental Appendix that forgone match is observed across characteristics of the plans. Supplemental Appendix Table D3 shows that

[17] See Supplemental Appendix A.4 for details on how we measure education and parental income.

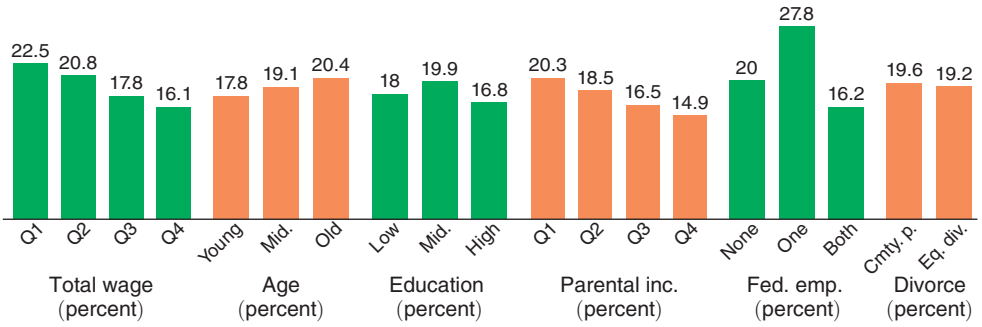


FIGURE 4. FORGONE MATCH AND DEMOGRAPHIC CHARACTERISTICS

Notes: Total wages are the total W-2 wages received by the couple (gross of pretax retirement contributions) at all jobs. We form quartiles. For age, couples are divided into three groups based on their mean age: Young (< 35), Middle ($35 \leq age < 50$), Older (≥ 50). Education groups are formed using calendar years in higher education (measured as receipt of Form 1098-T); low / medium / high corresponds to zero calendar years, one to four calendar years, and five or more calendar years, respectively. Parental income is divided in quartiles. See Supplemental Appendix A.4 for further details on the construction of parental income and education variables. We divide our sample by whether neither, one, or both spouses are federal employees. Finally, we divide the sample by whether the couple lives in a state governed by equitable division or community property divorce law. Education and parental income is only observed for a part of our sample. For the education analysis, we restrict to couples where both spouses attained age 18 in 1999 or later. For the parental income analysis, we restrict to couples where we are able to match each spouse to a parent. Forgone match is defined as the difference between (i) the match that would be received if total contributions were allocated to maximize the total match received and (ii) the match actually received.

forgone match varies only modestly across different types of retirement plans facing couples. Match schedules are piecewise linear functions, with most plans either having one or two tiers. We divide our sample of plans by whether there is no match, whether the match has one tier, or whether it has two (or more) tiers. The table reports results by the combination of plan types. The lowest incidence of forgone match (16.2 percent) is found among those where both spouses have single-tier plans, while the highest incidence (29.5 percent) is found among those where one spouse has a single-tier plan and the other has a two-or-more-tier plan. Average forgone match among those not coordinating is very similar across plans.

C. *Using Noncoordinating Benchmarks to Estimate the Share of Couples That Are Not Coordinating*

If a couple has positive forgone match ($FM > 0$), we can infer that they are not actively coordinating. However, the inverse is not true: No forgone match ($FM = 0$) does not imply that a couple is actively coordinating. Since the set of efficient allocations can be large, spouses could be making unilateral contribution decisions that just happen to be consistent with household efficiency. For example, both members of the couple could be fully exploiting their respective matches (in which case the marginal match rate would be zero for both spouses), or they could both face the same single-tier match schedule with neither saving beyond their match cap.

In this section, we estimate the share of couples not actively coordinating. Since actively coordinating couples have $FM = 0$, we can express the probability of

positive forgone match as the product of (i) the probability of not coordinating and (ii) the probability of positive forgone match conditional on not coordinating. We have directly observed that the probability of forgone match is 19.3 percent. Thus, given an estimate of the probability of positive forgone match conditional on not coordinating, we can estimate the share of couples not actively coordinating.

We implement two approaches to evaluate the probability of forgone match we would expect to see absent any coordination between spouses. First, we create placebo samples of synthetic couples with similar characteristics to real couples. Second, we observe couples in the years prior to marriage and the years following a divorce. Both approaches yield quantitatively similar results: In the absence of any coordination, we would expect to see in the region of 33 percent of couples allocate retirement contributions inefficiently. A comparison of the incidence of forgone match in the data with that in these no-coordination benchmarks suggests that about two-thirds of couples do not actively coordinate; that is, inefficiency is widespread.

Synthetic Couples Benchmarks.—For our first approach, we generate two placebo samples of synthetic couples for whom we should expect no coordination of retirement contributions. Formally, we denote our sample of couples as C, made up of couples $c_1, c_2, \ldots, c_N$. Each couple c_i is comprised of spouses a_i and b_i.[18] We generate two samples of synthetic couples. In the first one, we take the sample of married people in our data and rearrange them; we refer to this sample as the "Reshuffled Couples" sample. Specifically, for each a_i, we find a spouse b_i' from the set $B_i^{donor} = \left(b_1, b_2, \ldots, b_{i-1}, b_{i+1}, \ldots, b_N\right)$ such that b_i' matches the age, gender, and earnings of the true spouse b_i.[19] The couple $\hat{c}_i^M$ is then comprised of a_i and b_i'. In the second synthetic sample, we find a set of single individuals who match each individual (a_i and b_i) in our real sample; we refer to this sample as the "Pairs of Singles" sample. Let $\hat{A}_i$ denote the set of matches for a_i, which is comprised of single individuals $\hat{a}_1, \hat{a}_2, \ldots, \hat{a}_{N_i}$; $\hat{B}_i$ is defined similarly.[20] We then form all possible synthetic couples for couple i by taking the Cartesian product of $\hat{A}_i$ and $\hat{B}_i$. Within this set of potential synthetic couples, we choose the couple $\hat{c}_i^S = (a_i'', b_i'')$, whose total DC contributions match the total DC contributions of the true couple most closely.

As in our actual sample, we observe the incentives and contributions of each synthetic spouse in our two placebo samples. We can therefore calculate forgone match for each couple in the synthetic samples. We use the incidence of forgone match in the synthetic samples as estimates of the (counterfactual) incidence of *FM* that would be observed in the absence of any coordination.

Columns 1 and 2 of Table 3 compare our observed forgone match to the values in the two noncoordination benchmarks. Panel A reports the observed share with forgone match in the baseline sample of true couples. Panel B gives the proportion with forgone match in each of the two synthetic samples. These estimates imply that,

[18] Our sample includes both different-gender and same-gender couples (as well as a small number of couples where at least one member has unknown gender). We randomly assign one member of the couple to be *a* and the other *b*.

[19] We enforce an exact match on gender and year of birth. Within the set of possible matches, we choose the b_i' that matches the closest on earnings.

[20] We require an exact match on firm and gender, a match in age within ten years, and a match in earnings within 20 percent. For computational tractability, we require that $\hat{A}_i$ and $\hat{B}_i$ contain no more than ten individuals; if there are more than ten satisfying the match conditions, we choose the ten with the closest match on earnings.

TABLE 3—ESTIMATED PROPORTION FAILING TO ACTIVELY COORDINATE

	Baseline sample		Four years after marriage	Four years before divorce
	(1)	(2)	(3)	(4)
Panel A. Sample of true couples				
Proportion with forgone match	0.193		0.177	0.201
	Reshuffled couples	Pairs of singles	Four years before marriage	Four years after divorce
Panel B. Noncoordination benchmark				
Proportion with forgone match	0.331	0.336	0.315	0.339
Panel C. Implied share noncoord.				
	0.582	0.574	0.562	0.595

Notes: Panel A reports the share of couples that have forgone match in each sample. In columns 1 and 2, we use the baseline sample. In column 3, we use our panel observations four years after their marriage in the sample from panel A of Figure 5. In column 4, we use our panel observations four years prior to their observed divorce in the sample from panel B of Figure 5. Panel B reports the benchmark estimate—that is, the estimate for the probability of observing forgone match when not coordinating. In columns 1 and 2, the benchmarks are the Reshuffled couples and Pairs of singles samples, respectively. The former is constructed by taking the married couples in our sample and rearranging them. A (random) reference spouse in each couple is matched with someone with the same age and gender and similar earnings as their actual spouse. The latter sample is formed by arranged singles in synthetic couples such that the combination of ages, gender, earnings, and the total contributions match the real couple. In column 3, the benchmark is the same set of couples as in column 3 of panel A, but four years prior to their marriage. In column 4, the benchmark is the same set of couples as in column 4 of panel A, but four years after their divorce. In panel C, we report the implied share of true couples that are not coordinating in each sample, based on the benchmark in panel B, equal to the proportion with $FM > 0$ in the true sample (i.e., panel A) divided by the proportion with $FM > 0$ in the benchmark in question (i.e., panel B). Forgone match is defined as the difference between (i) the match that would be received if total contributions were allocated to maximize the total match received and (ii) the match actually received.

without any coordination within the household, 33 percent to 34 percent of placebo couples end up with some forgone match. Panel C uses these two figures to estimate that share of couples not coordinating is 57 percent to 58 percent.[21]

Evolution around Marriage and Divorce.—Our second noncoordination benchmark leverages the fact that we have longitudinal data. We calculate the share of couples who have forgone match at their chosen saving choices prior to their marriage and/or after their divorce. Figure 5 shows the probability of having some forgone match in a balanced panel four years prior and after marriage and divorce. Panel A shows that in the nine-year period centered around the time of marriage, the incidence of noncoordination falls by 14 percentage points. Panel B shows that in the nine-year period centered around divorce, the incidence of noncoordination increases by 14 percentage points. In the case of marriage, the trend starts several years before marriage, indicating that the tendency toward coordination (for those who coordinate) is a gradual process. In the case of divorce, the unwinding of cooperation starts just before the year of divorce.

[21] Formally, some couples are coordinating ($Coord = 1$) and some are not ($Coord = 0$). Those who are coordinating have $FM = 0$; those who are not have $FM > 0$, with some probability $\Pr(FM > 0 \mid Coord = 0)$. Thus, we can write $\Pr(FM > 0) = \Pr(FM > 0 \mid Coord = 0)\Pr(Coord = 0)$. Since we observe $\Pr(FM > 0)$, the estimates of $\Pr(FM > 0 \mid Coord = 0)$ (from our synthetic samples) allow us to recover $\Pr(Coord = 0)$, equal to $\Pr(FM > 0)/\Pr(FM > 0 \mid Coord = 0)$.

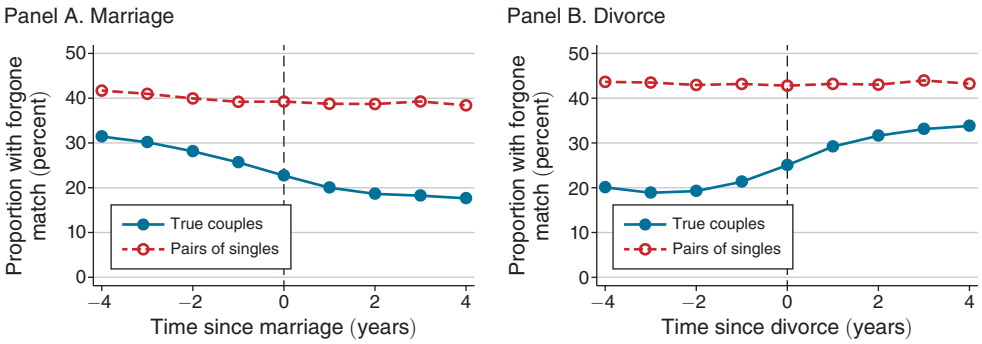


FIGURE 5. PROBABILITY OF NONCOORDINATION AROUND MARRIAGE AND DIVORCE

Notes: These graphs (solid series) show the probability of noncoordination around marriage (panel A) and divorce (panel B) where the dependent variable is having some forgone match (over a de minimis threshold of $10) using our panel data. The sample is comprised of all individuals in our panel restricted to those for whom we have at least four years of consecutive data on each side of the event in question. We require that all couple-year observations satisfy our baseline vesting, tenure, and age restrictions. The dashed series show the probability of noncoordination for a sample of synthetic couples, comprised of single individuals, constructed in a manner analogous to the $\hat{C}_S$ sample in the cross section. Forgone match is defined as the difference between (i) the match that would be received if total contributions were allocated to maximize the total match received and (ii) the match actually received.

In both panels, we include a panel of synthetic couples as well. These samples are analogous to the $\hat{C}_S$ sample in the cross section: We find each spouse an unmarried individual with similar age, similar earnings, same gender, and who works at the same firm. We form all such possible couples, choosing the one with total contributions that match the true couple most closely.[22] These synthetic couples do not experience any substantial change in their probability of forgoing a match, suggesting that the patterns in the true couples around marriage and divorce are not spuriously driven by aging or any other factors.[23]

In both cases, the noncooperative benchmark (four years premarriage and four years postdivorce for true couples) is that approximately 32 percent to 34 percent of couples would have some forgone match. Columns 3–4 of Table 3 use these benchmarks to infer the share of true couples not actively coordinating. Panel A contains the observed share forgone match for true couples in each sample.[24] For those we observe marrying (divorcing), we measure forgone match four years after (before) marriage (divorce). Panel B of Table 3 reports the share of forgone match in our noncoordination benchmarks (i.e., four years before marriage or four years after divorce). Finally, panel C compares true couples to these noncoordination benchmarks. The implied share of couples not coordinating when comparing the same couples before / after divorce and marriage is close to that we estimate when

[22] The procedure is nearly identical to the $\hat{C}_S$ sample in the cross section. The only exception is that earnings and contributions are computed as the average of the nine-year period surrounding the divorce or marriage event.

[23] In Supplemental Appendix D.5, we study marriage and divorce in an event study framework, leveraging the synthetic couples as never-treated units. These results are qualitatively similar (with slight attenuation) when controlling for total contributions and earnings.

[24] The samples here are different from our baseline and are comprised only of couples where the event (marriage or divorce) is observed and for whom we have four years of consecutive data on each side of the event in question.

comparing our baseline sample to our synthetic benchmarks of random pairs with similar characteristics to real couples.

IV. What Explains Lack of Coordination?

A. *Supplementing the Administrative Data with Survey Evidence*

In this section, we examine the mechanisms behind the lack of coordination that we have documented. We use both our administrative data and a tailored survey, which we ran through the survey platform Prolific. We surveyed 1,000 individuals screened to be living in the United States, working, married, aged between 18 and 59, and who had a retirement plan. Supplemental Appendix E describes the survey design and provides a stand-alone discussion of the results.The survey data is available in the repository for Choukhmane et al. (2025).

Central to the survey was a hypothetical choice problem in which respondents were asked to divide a given level of savings between their account and their spouse's account, mimicking the maximization problem faced by couples in reality. The two accounts had different match schedules. We varied the combinations of match schedules such that there were three versions, which differed in the match-maximizing strategy. For one-third of respondents, the match-maximizing strategy was to put all the funds in their own account, for one-third it was to put all the funds in their spouse's account, and in the final third it was to split it between accounts. We also measure respondents' financial literacy, their understanding of the rules around asset division in divorce, and details about their retirement plans and those of their spouse.

We find that 40 percent of respondents have forgone match; that is, they allocate their savings in a way that does not maximize the amount of employer match.[25] Subsequent questions allow us to distinguish between whether the forgone match was deliberate or not. We inform respondents when their allocation did not maximize the match, ask them whether this was intentional, and if it was not, we give them an opportunity to make a new choice. We classify those who did so intentionally as well as those who didn't realize but declined to reoptimize as having "deliberate" forgone match. Those who chose to reallocate and whose subsequent choice has no forgone match are classified as having *accidental* forgone match.[26] Of the 40 percent of survey respondents who have some forgone match, approximately 45 percent have accidental forgone match and 48 percent have deliberate forgone match, with the remaining 7 percent having "other" or unknown forgone match.[27]

[25] For several reasons, we expect the incidence of forgone match to differ between the survey and the administrative data. In the survey, only one allocation achieves efficiency by construction, while in the administrative data, there is often a range of allocations consistent with efficiency. Second, the intrahousehold dimension of the problem is more salient in the hypothetical survey experiment. Third, unlike in the administrative data, survey respondents effectively act as dictators, choosing the contribution to both accounts. Nevertheless, we expect that the correlates of forgone match in the survey data can be informative about the drivers of forgone match for real couples.

[26] A third possibility is that the respondent makes a new choice but that new choice also has forgone match. In this small number of cases (39), we subjectively classify the respondent into "accidental," "deliberate," or "other" (i.e., unknown) forgone match based on narrative explanations. Supplemental Appendix E.4.2 gives further details.

[27] "Other" forgone match comprises (i) those who did not understand the question about whether they realized that they had not maximized the match and (ii) those who opted to reallocate, did not maximize the match, but whose narrative explanation was not sufficient to determine a classification.

In the next two sections, we further investigate drivers of noncoordination, taking insights both from the administrative data and the survey evidence. We segment our investigation into noncoordination as a mistake (in Section IVB) or as a deliberate decision (in Section IVC).

B. *Noncoordination as a Financial Mistake*

We start by showing evidence against some factors that might have driven our result: We use our administrative data to show that noncoordination is not substantially driven by inertia or by heuristics that lead spouses to contribute equal amounts. We then show that the incidence of forgone match does not vary with the financial stakes of the match, suggesting that rational inattention is not driving the results. We then use our survey data to show, though, that inattention (albeit perhaps not of a rational type) is likely relevant: Many couples have not even considered that there might be gains to coordination, and when the implications of their decisions are brought to their attention, they change their behavior.

Noncoordination Is Not Driven by Inertia or Auto-Enrollment.—A plausible explanation for noncoordination could be inertia: Couples may want to maximize their match but fail to do so due to passive behavior or adjustment costs. Indeed, inertia has been shown to be consequential in the retirement saving setting (Madrian and Shea 2001; Derby, Mackie, and Mortenson 2023; Choukhmane 2024). To investigate whether inertia can account for the observed lack of coordination across spouses, we exploit the longitudinal aspect of our data. To the extent that inertia is driving our results, we would expect that when couples make an *active* change to their contributions, they should eliminate forgone match. We focus on the subsample of couples where both spouses made an active change to their retirement plan contributions, and we evaluate the *change* in the extent of their forgone match. In defining this sample, we further restrict to those with plans that did not have auto-features, to ensure that the changes in the contributions that we observe arise from active decisions.

Supplemental Appendix Figure D1 shows the distribution of the *change* in the quantity of forgone match by couples in which both spouses made an active change in their contribution rate (we omit a large mass at zero). This distribution has a mean of minus $11—that is, the mean change in forgone match after an active change was very small. Further, the distribution is close to symmetric; the share of couples *reducing* the extent of their forgone match (9.4 percent) is only slightly larger than the share of couples *increasing* it (8.1 percent). This suggests that the noncoordination we document is not driven by spouses being temporarily away from the efficient allocation due to inertia.

A second manifestation of behavior driven by inertia would be if those individuals in plans that offer auto-enrollment stay at the auto-enrollment default. There is, of course, no reason that the default contribution rates should align with the contribution rates that maximize employer matching at the household level. However, we find the incidence of forgone match among couples not subject to auto-enrollment is similar to the share of inefficient allocations in the baseline sample: 18.8 percent for couples hired under an opt-in regime relative to 19.3 percent in the overall sample.

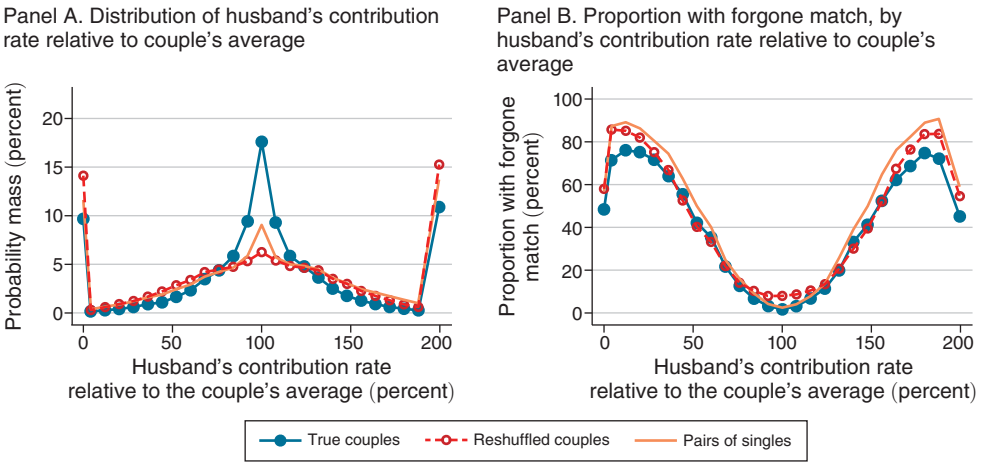


FIGURE 6. EQUAL SAVING HEURISTICS: DENSITY, AND PROBABILITY OF FORGONE MATCH, BY HUSBAND'S SHARE OF CONTRIBUTION

Notes: Panel A plots the density of the husband's contribution rate relative to the couple's average contribution rate, that is, $\frac{s^H/y^H}{0.5\left(s^H/y^H + s^W/y^W\right)}$. Panel B plots the probability of forgone match as a function of the husband's contribution rate relative to the couple's average contribution rate, calculated in the same manner. We drop couples where at least one spouse is contributing greater than 95 percent of the statutory maximum on individual contributions ($18,000 or $24,000 depending on age). This figure restricts to different-gender couples. Forgone match is defined as the difference between (i) the match that would be received if total contributions were allocated to maximize the total match received and (ii) the match actually received.

We conclude that inertia is not a major driver of the patterns of noncoordination.

Equal-Saving Heuristics Do Not Explain Inefficient Allocations.—Next, we investigate whether couples allocate contributions to retirement accounts in a manner that is "equal" but does not consider economic incentives at the household level.[28] Couples may fail to allocate their savings efficiently because they abide by a rule of contributing equal amounts to their respective retirement accounts. We investigate this hypothesis in Figure 6. In panel A, we plot the density of couples as a function of the husband's contribution rate relative to the couple's average contribution rate, both for the true sample and the two synthetic samples.[29] More precisely, we first calculate the contribution rate of each spouse $\left(s^j/y^j\right)$ for $j = \{H, W\}$ (husband and wife). We then measure how they relate to each other by calculating a "husband's relative contribution" as follows: $\frac{s^H/y^H}{0.5\left(s^H/y^H + s^W/y^W\right)}$. This quantity ranges from 0 percent to 200 percent, and a relative contribution of 100 percent indicates that both spouses are contributing the same proportion of their salary. We

[28] Gathergood et al. (2019) show that heuristic-type behavior explains how individuals allocate credit card repayments across cards.

[29] For this exercise, we drop same-gender couples. We also drop couples where at least one spouse is contributing 95 percent or greater of the statutory maximum, which is $18,000 or $24,000 depending on age. This second restriction eliminates couples whose savings are equal to each other because they are both contributing at the statutory maximum.

find that, indeed, there is an excess mass near 100 percent. This finding, which to our knowledge is novel, is consistent with some couples engaging in an equal-saving heuristic.[30] In Supplemental Appendix Figure D2, we show the figures are qualitatively similar if the husband's relative contribution is measured in dollars rather than proportions of salary, that is, $\frac{s_H}{0.5(s_H + s_W)}$.

However, this type of behavior seems to, if anything, attenuate, rather than contribute to, the prevalence of inefficient allocations that we document. Panel B of Figure 6 plots the share of couples with *FM* as a function of the husband's contribution share. In both the true and synthetic samples, the proportion with *FM* is *smaller* when the husband's relative contribution is near 100 percent. Mechanically, this occurs because such an allocation is less likely to involve one spouse contributing beyond their match cap while the other fails to exploit their match. Thus, if anything, equal-saving heuristics may be *reducing*, rather than creating, forgone match.

Noncoordination Persists When Stakes Are High.—If the cost of failing to coordinate is sufficiently low, then *rational* inattention could explain the noncoordination that we observe. To test this hypothesis, we use both spouses' matching formulas to construct a measure of "stakes" by asking the following question: Given the couples' combined savings and their combination of match schedules, what is the maximum match they could forgo from not coordinating saving decisions? For some couples, stakes are lower because the worst-case outcome from not coordinating contributions generates only a small (or zero) amount of forgone match. In other cases, noncoordination could cost several thousand dollars in forgone match. To do this, we first calculate the maximum match and minimum match a couple can obtain at their combined savings. The minimum match might be zero (if one spouse is in an employer that does not offer any match) or positive if both spouses have a match. The difference between the match the couples receive in the best case and that which they receive in the worst case is our measure of the "stakes" of the decision. Figure 7 shows how forgone match varies with this measure of the stakes. In panel A we measure the stakes in dollars, while in panel B we express the stakes as a proportion of the joint earnings of the couple. In each panel, the solid blue bars represent true couples, while the lighter red and orange bars represent the two samples of synthetic couples. We take two lessons from this graph. First, whether the stakes are low or high, true couples outperform synthetic couples; that is, some coordination occurs. But, second, even when the stakes are extremely high (e.g., when noncoordination could cost the couple more than $5,000 or 5 percent of their earnings per year), a substantial proportion of couples are allocating their saving inefficiently, and whether we measure stakes in dollars or as a share of earnings, the incidence of coordination is not sensitive to stakes of the decision.

While we take this as evidence that rational inattention is not driving our results, in the next subsection we show that inattention (albeit perhaps not of a rational type) *is* a contributing factor.

[30] A contributing factor to this result could be a tendency for spouses to have equal earnings (Bertrand, Kamenica, and Pan 2015). However, even among spouses whose earnings differ, we find evidence of a tendency for spouses to have similar saving rates.

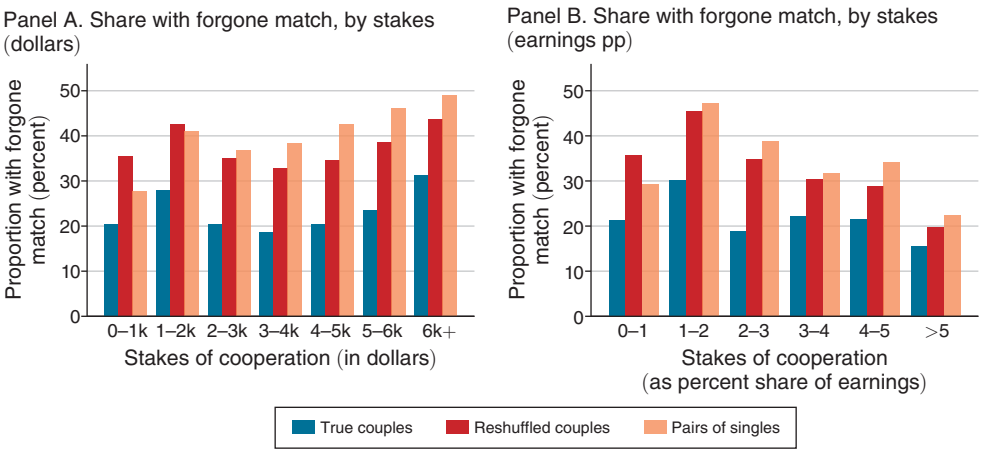


FIGURE 7. SHARE WITH FORGONE MATCH AS A FUNCTION OF THE "STAKES" OF THE DECISION

Notes: This figure plots the share of couples with forgone match as a function of the "stakes" of the decision. The "stakes" are defined as the difference between the maximum possible match and the minimum possible match, given the total contributions of the couple and the match schedule faced by each spouse. In panel A, the stakes are measured in dollars. In panel B, the stakes are scaled by the total earnings of the couple. This figure uses the baseline sample. Forgone match is defined as the difference between (i) the match that would be received if total contributions were allocated to maximize the total match received and (ii) the match actually received.

Many Couples Have Not Considered There Might Be Gains to Coordination.—We provide two pieces of evidence from our survey data that many couples might not have considered that there can be gains to coordination.

First, for the respondents who (i) had forgone match in our hypothetical choice experiment and (ii) had stated that they had not realized this fact, we asked them whether they would like to change their answers; 90 percent stated they would like to change their allocation, and of those, 80 percent reallocated their contributions to remove forgone match (details are given in Supplemental Appendix Table E3). These respondents had not noticed the intrahousehold arbitrage opportunity available to them, despite the fact that, in our survey, this fact was surely more salient than it would have been in reality. When prompted to reoptimize and offered the chance to, most took the opportunity.

Second, for the subsample of our survey respondents where both spouses have a DC plan, we explicitly asked respondents whether they had considered that there might be gains from coordination; 36 percent of this subsample, and 55 percent of those with accidental forgone match, answered that they had not considered this prospect (Supplemental Appendix Table E9 shows these results). This stated lack of attention to the potential arbitrage opportunity in their real retirement savings decisions correlates with a demonstrated lack of attention to the hypothetical choice experiment. Those who state that they have considered the gains to coordinating in reality were less than half as likely to have accidental forgone match in the hypothetical survey experiment than those who had not considered it (with, respectively, 13 percent and 29 percent of each group having accidental forgone match).

The salience of this arbitrage opportunity at the household level might be limited because retirement plans are individually managed and sponsored by one's

Table 4—Forgone Match and Its Types in the Survey Data

	Observations (1)	Prop. of total (2)	Prop. of those w / forgone match (3)
No forgone match	594	60.2	
Has forgone match	393	39.8	
Of which			
… Accidental forgone match	177	17.9%	45.0%
… Deliberate forgone match	187	18.9%	47.6%
… Other	29	2.9%	7.4%

Notes: The sample for this table is the full survey sample. The first two rows give number and proportion of observations by whether they have forgone match or not. The last three rows give number and proportion of observations by the type of forgone match. The proportions are given out of the total sample (column 2) and out of just those with forgone match (column 3). A full description of our classification of forgone match into its types is given in Supplemental Appendix E.4.

employer. The household dimension of the decision is potentially more salient for couples working for the same firm. Consistent with this, Table 5 shows that forgone match is less common for couples who work for the same employer, even compared to those who face identical match schedules but work at different employers.

Noncoordination Falls with Financial Literacy.—A growing literature emphasizes financial literacy's role in financial decision-making (see Lusardi and Mitchell 2023 for a recent survey). Couples with lower levels of financial literacy might be more prone to financial mistakes and less likely to realize that coordinating retirement contributions can lead to their receiving larger matching contributions. We measure financial literacy based on the responses to five questions. Four questions are taken from the existing literature and capture respondent's understanding of simple financial concepts (we take three questions taken from Lusardi and Mitchell 2011 and one from Lusardi 2008). The fifth question is new and tailored to our setting: We survey respondents on their knowledge of the tax treatment of traditional 401(k) accounts. The questions are given in Supplemental Appendix E.2.4. We group respondents by whether they answered two or fewer, three, four, or five questions correctly.[31]

Table 6 summarizes the relationship between financial literacy and forgone match. Column 1 gives the proportion of our sample in each of the financial literacy groups. Columns 2 to 5 show incidence of forgone match overall and for each type. Forgone match falls steeply with measured financial literacy.[32]

Taking Stock.—We take these results as evidence that a sizable share of noncoordination is a result of a financial mistakes. Inertia or a heuristic of equal

[31] Only approximately 5 percent of respondents got 1 or no questions correct, hence our grouping at the bottom.

[32] This gradient might partly reflect differences in survey engagement: Respondents who are less engaged in the survey are presumably more likely to get financial literacy questions wrong and to choose a savings allocation that does not maximize the match.

Table 5—Forgone Match for Those Working at the Same and Different Firms

Circumstance	Proportion with forgone match (%)
Different match, different firm	29.2
Same match, different firm	25.4
Same match, same firm	15.2

Notes: "Same match" means that both spouses face a schedule with identical match rates and match caps (expressed as a share of earnings). Forgone match is defined as the difference between (i) the match that would be received if total contributions were allocated to maximize the total match received and (ii) the match actually received.

saving do not appear to drive our results; a combination of inattention to the fact that there might be gains from coordination and financial literacy does.

C. *Noncoordination as a Deliberate Decision*

Table 4 showed that approximately half of those who did not maximize the match in response to our hypothetical choice question stated that they did so deliberately. Section ID discussed reasons why spouses might have (or perceive that they have) incentives to prefer contributions to their own accounts and that such reasons might be correlated with the degree of commitment. In this subsection we provide evidence that this deliberate decision to forgo match is associated with proxies for the strength of marital commitment, is more prevalent for those who believe that they will keep their own accounts on divorce, and that notions of trust in one's spouse and fairness contribute to the decision.

Noncoordination Is Correlated with Proxies for Commitment.—We propose five measures in our administrative data that are plausibly correlated with the strength of marital commitment. These are the length of the marriage, the presence of children, the couple having a mortgage, a divorce event in the near future,[33] and whether the couple had a joint bank account before getting married. For the first four of these, we observe the indicator for our entire sample. For the last—whether they had a joint back account before marriage—we can only construct if the couple was first observed as married in 2008 or later.[34]

Table 7 shows the relationship between these proxies of marital commitment and the occurrence of forgone match (panel A) and the size of the forgone match as a proportion of employee contributions (panel B). We can construct our joint bank account indicator for only a subset of our data, and so columns 1 and 3 omit that and use our full sample, while columns 2 and 4 include it and use the sample for which it is observed. We control for household earnings, household contributions and their

[33] The results we show are for a cross section in 2015, and as 2022 represents the most recent year of tax data, our divorce indicator will only capture divorce realizations in the seven years subsequent to the year of observation.

[34] We code them as using a joint bank account if each member of the couple used the same bank account to receive a direct deposit of any tax (Form 1040) refund that they are entitled to. For this analysis, we restrict attention to those couples where both members of the couple received a Form 1040 refund in the final year prior to marriage and both elected direct deposit.

Table 6—Financial Literacy

	Prop. of sample (1)	Prop. w / any forgone match (2)	Prop. w / deliberate forgone match (3)	Prop. w / accidental forgone match (4)	Prop. w / other forgone match (5)
≤ 2	17.4	64.0	29.1	29.1	5.8
3	22.1	49.1	27.1	18.8	3.2
4	36.9	32.1	14.6	15.9	1.6
5	23.6	25.3	10.7	12.0	2.6

Notes: This table groups our survey respondents by the number of financial literacy questions they got correct. Supplemental Appendix E.2.4 gives the questions. Column 1 gives the proportion of the sample getting each number of questions correct. The remaining columns show, by the number of questions correct, the proportion with forgone match. Column 2 gives proportion with any forgone match; columns 3, 4, and 5 give proportion with deliberate, accidental, and other forgone match, respectively.

interaction, and a set of other characteristics, listed in the table notes. Supplemental Appendix Tables D5 and D6 report the coefficients on these characteristics.

In each case we see that those who are less committed (per our proxies) are more likely to have forgone match. The magnitudes of the association between some of these proxy variables and noncoordination is sizable, especially given that they are likely to be only noisy measures of the strength of marital commitment. Having a mortgage, for example, is associated with a 2.4 percentage point reduction in noncoordination (relative to an overall mean of 19.3 percent) in our specification with full controls. The imminence of divorce has an association of a roughly similar magnitude but of the opposite sign. Those with a joint account are 1.5 percentage points less likely to have some forgone match (from a base of 18.3 percent in this sample).

While these correlations are not conclusive evidence, and these noisy proxies only explain a small share of the total variation in the data, we take them as suggestive that spouses more committed to each other are, all else equal, less likely to forgo match. This raises a question. Even in the absence of commitment to the marriage, if divorce is the outside option, spouses should still face strong incentives to coordinate, raising the question as to why there is a link between commitment and cooperation. The next subsection uses our survey to discuss one possible reason for this.

Knowledge of Divorce Law Is Imperfect and Correlated with Coordination.—In our survey, we measured perceptions of how retirement wealth and other assets are treated in divorce (Supplemental Appendix E.4.5 provides details). Wealth in a retirement account is a marital asset and, as such, is divided on divorce independent of who made the contributions.[35] Despite this, as column 1 of Table 8 shows, over one-third of couples believe that they would keep their own retirement wealth, and close to one-fifth do not know how assets would be divided. Around half of respondents say that they would split the assets or that they would be divided in some other manner.

[35] States differ between those that are "community property" states, in which assets acquired during marriage tend to be divided equally, and "equitable division" states, in which there is some judicial discretion to the splitting of assets. However, even in equitable division states, the identity of the particular spouse who made the retirement contributions should not be relevant to the division of retirement assets.

TABLE 7—FORGONE MATCH AND COMMITMENT

	Prop. with forgone match		Forgone match as prop. of emp'ee contribution	
	(1)	(2)	(3)	(4)
Length of marriage	−0.0010	−0.0019	−0.0141	−0.0292
	(0.0002)	(0.0006)	(0.0039)	(0.0117)
Kids	−0.0057	−0.0110	−0.1975	−0.1764
	(0.0020)	(0.0039)	(0.0401)	(0.0740)
Future divorce	0.0181	0.0108	0.2416	0.1192
	(0.0031)	(0.0054)	(0.0679)	(0.1058)
Mortgage	−0.0244	−0.0321	−0.3487	−0.4088
	(0.0025)	(0.0055)	(0.0541)	(0.1097)
Joint account prior to marriage		−0.0151		−0.2849
		(0.0053)		(0.0990)
Baseline mean	0.1929	0.1830	2.4302	2.1510
Partial R^2	0.0011	0.0021	0.0007	0.0012
Inc. × conts. controls	X	X	X	X
Full controls	X	X	X	X
Observations	184,570	44,440	184,570	44,440

Notes: The dependent variable in columns 1 and 2 is an indicator variable for a couple having some forgone match, and the coefficients are those from a linear probability model. The dependent variable in columns 3 and 4 is the forgone match as a proportion of total employee contributions, scaled in percentage points; the coefficients are those from OLS regressions. All rows control for the interaction of total earnings and total contributions by the couple, where earnings and contributions are measured with respect to employers $j(i_A)$ and $j(i_B)$ only. Our full set of controls additionally includes the mean age of spouses, the age gap between them, the mean tenure of the couple, tenure gap between them, the share of earnings earned by the primary earner, whether one or both members of the couple was hired during an automatic enrollment regime, whether they live in a state where divorce law requires equitable division, the log of adjusted gross income, whether each member of the couple faces an identical match formula, and whether both members of the couple work for the same firm. The full set of coefficients for the probability of noncoordination and the extent of noncoordination are reported in Supplemental Appendix Tables D5 and D6, respectively. Columns 1 and 2 use our baseline sample. Columns 3 and 4 use the same for whom we can ascertain whether they had a joint bank account before marriage.

Columns 2, 3, 4, and 5, show, conditional on a particular answer to the question, what proportion of respondents have any forgone match, deliberate forgone match, accidental forgone match, or other forgone match, respectively. There is a clear pattern. Those who think that accumulated retirement wealth remains one's own property on divorce are much more likely than those who think such wealth with be split to have forgone match. The difference comes largely from a tendency to have substantially greater deliberate forgone match. It is clear that, for a share of the population, placing wealth in one's own account is viewed as a device to protect one's own wealth in case of divorce.

The Role of Gender.—Across the administrative and survey samples, we find evidence consistent with women placing a higher value on maintaining their own retirement savings than men do.

In each couple that is not coordinating, efficiency would involve reallocating contributions from an oversaving spouse to an undersaving spouse. Among different-gender couples with some forgone match in the administrative data,

Table 8—Knowledge of Divorce Law and Association with Forgone Match (Percent)

	Prop. of sample (1)	Prop. w / any forgone match (2)	Prop. w / deliberate forgone match (3)	Prop. w / accidental forgone match (4)	Prop. w / other forgone match (5)
Keep own	34.2	51.2	27.8	19.2	4.1
Split / other	46.9	36.9	15.3	19.0	2.6
Don't know	18.8	26.3	11.8	12.9	1.6

Notes: This table groups our survey respondents by their knowledge of divorce law. Supplemental Appendix E.4.5 describes the construction of the groups. Column 1 gives the proportion in each group. The remaining columns summarize, by divorce knowledge, the incidence of forgone match. Column 2 gives proportion with any forgone match; columns 3, 4, and 5 give proportion with deliberate, accidental, and other forgone match, respectively.

Figure 8 shows how the probability of being the oversaver varies with their household share of earnings and their gender. The probability of being the oversaver increases with share of earnings. More strikingly, throughout the relative earnings distribution, wives are more likely to be the oversaving spouse relative to husbands. This finding suggests that women put a higher value on maintaining their own retirement savings.

This hypothesis is also consistent with the survey's results. As shown in Supplemental Appendix Table E6, among respondents who received the survey question where maximizing the match requires putting all the savings in the spouse's account, female respondents are more likely to have deliberate forgone match than male respondents (31.5 percent of women and 24.7 percent of men chose deliberate forgone match in this scenario). In contrast, when achieving efficiency does not require giving up control over the savings, we find little differences in deliberate forgone match by gender (i.e., 14.3 percent of women and 14.1 percent of men chose deliberate forgone match when the efficient allocation requires putting all the savings in their own account). We find similar results for different-gender couples in the administrative data. When it is optimal to save entirely in the wife's account, 38.7 percent of husbands nevertheless make positive contributions; when it is optimal to save entirely in the husband's account, 47.1 percent of wives nevertheless make positive contributions.

These differences across genders could reflect differences in beliefs about divorce outcomes. In the survey, women were more likely to believe retirement assets would be kept separate in case of a divorce: 39.6 percent of women and 28.9 percent of men expressed this incorrect understanding of divorce law.

The Role of Ownership of Accounts, Fairness, and Risk.—We conclude with a brief discussion of narrative explanations given in our survey for deliberate noncoordination. We asked all survey respondents who had forgone match (and did not correct their allocation later on) for a narrative explanation of their choice. From these narratives, we identified three main themes present in close to two-thirds of the responses: (i) the desire for financial independence, (ii) considerations of fairness, and (iii) concerns about risk. First, many respondents indicated a sense of personal ownership over their own account with a corresponding lack of ownership over their spouse's account. For instance, one respondent wrote, "I need independence. I don't



FIGURE 8. DIFFERENCES BY RELATIVE EARNINGS AND GENDER

Notes: For each couple with forgone match, efficiency can be attained if savings are reallocated from one spouse (the "oversaver") to the other (the "undersaver"). This figure restricts to different-gender couples with forgone match. It plots the share of individuals who are oversavers as a function of the share of the couple's earnings that that individual represents, separately for wives (solid circles) and husbands (hollow circles). For legibility, we restrict to cells where the earnings share ranges from 15 percent to 85 percent. Forgone match is defined as the difference between (i) the match that would be received if total contributions were allocated to maximize the total match received and (ii) the match actually received.

want to put my money in someone else's account … even if it is my husband." Another one wrote, "Our finances are not combined. We each may have different retirement needs and trajectories." In some cases, this preference for financial independence was associated with a lack of trust in ones' spouse. For instance, one respondent wrote, "My husband has bad spending habits, so I would rather have things in my account." Second, several respondents cited fairness as a key consideration when making these allocation decisions. Many respondents said they aimed for an equitable distribution of funds, regardless of the potential for higher employer matches. For instance, one respondent wrote, "I wanted to be fair and split the money even though my spouse's employer matches more money." Third, many responses highlighted the importance of risk, with some specifically addressing the risk of divorce, while others pointed to general concerns about the future. One respondent wrote, "I wanted to ensure that in the event of any issues such as divorce or death, each of us had money. I would rather hedge against this and pay a fee of lost money than have none at all if unforeseen circumstances happen." Another one wrote, "You never know what's going to happen, so you don't want to put all the money in your spouse's retirement account." Although interpreting these responses involves some degree of subjectivity, the lesson we take is that for those couples who deliberately chose to forgo some employer match, a substantial driver is that respondents don't view their and their spouse's retirement accounts as interchangeable.

V. Conclusion

We show that many married couples fail to take full advantage of arbitrage opportunities available at the household level. Exploiting differences in matching incen-

tives across employers, we find that one-fifth of couples could increase their total retirement saving, by an average of over $700 per year, simply by reallocating some of their existing contributions to the account of the spouse with a higher marginal employer match rate. In the absence of any coordination, we estimate that the proportion of couples who could similarly increase their saving would be around 33 percent.

We show, by complementing our administrative data with a survey, that part of this behavior is accidental, driven by the fact that couples do not appreciate that there are potential gains from coordination, and that part is deliberate, driven in part by a desire for control and in part by an (often erroneous) belief that one's 401(k) remains exclusively one's own property in retirement.

Models where couples fail to realize the surplus available to them have been proposed; for example, the models of Basu (2006) and Hertzberg (2024) have the implication that spouses rely on inefficient strategies, and in the models of Lundberg and Pollak (1993); Browning, Chiappori, and Lechene (2010); and Del Boca and Flinn (2012), spouses can behave efficiently but subject to a noncooperative and potentially inefficient threat point. These alternatives to the workhorse collective household model are, however, used less frequently in applied work. We take our results as suggesting a greater role for noncooperative models in the study of households' economic decisions.

REFERENCES

Adams, Abi, Laurens Cherchye, Bram De Rock, and Ewout Verriest. 2014. "Consume Now or Later? Time Inconsistency, Collective Choice, and Revealed Preference." *American Economic Review* 104 (12): 4147–83.

Addoum, Jawad M. 2017. "Household Portfolio Choice and Retirement." *Review of Economics and Statistics* 99 (5): 870–83.

Alexander, J. Trent, David Bleckley, Jonathan Fisher, Katie Genadek, Susan Hautaniemi Leonard, and Aristotle Magganas. 2024. "Newly Available Individual-Level US Tax Data from 1969–1994." ADEP Working Paper 2024-02.

Almås, Ingvild, Alex Armand, Orazio Attanasio, and Pedro Carneiro. 2018. "Measuring and Changing Control: Women's Empowerment and Targeted Transfers." *Economic Journal* 128 (612): F609–39.

Andersen, Steffen, John Y. Campbell, Kasper Meisner Nielsen, and Tarun Ramadorai. 2020. "Sources of Inaction in Household Finance: Evidence from the Danish Mortgage Market." *American Economic Review* 110 (10): 3184–3230.

Arnoud, Antoine, Taha Choukhmane, Jorge Colmenares, Cormac O'Dea, and Aneesha Parvathaneni. 2021. "The Evolution of US Firms' Retirement Plan Offerings: Evidence from a New Panel Data Set." NBER Center Paper NB20-14.

Ashraf, Nava. 2009. "Spousal Control and Intra-household Decision Making: An Experimental Study in the Philippines." *American Economic Review* 99 (4): 1245–77.

Attanasio, Orazio P., and Valérie Lechene. 2014. "Efficient Responses to Targeted Cash Transfers." *Journal of Political Economy* 122 (1): 178–222.

Aura, Saku. 2005. "Does the Balance of Power within a Family Matter? The Case of the Retirement Equity Act." *Journal of Public Economics* 89 (9-10): 1699–1717.

Bargain, Olivier, Damien Echevin, Audrey Etienne, Nicolas Moreau, and Adrien Pacifico. 2022. "Tax Minimization by French Cohabiting Couples." *National Tax Journal* 75 (2): 265–96.

Basu, Kaushik. 2006. "Gender and Say: A Model of Household Behaviour with Endogenously Determined Balance of Power." *Economic Journal* 116 (511): 558–80.

Bergstresser, Daniel, and James Poterba. 2004. "Asset Allocation and Asset Location: Household Evidence from the Survey of Consumer Finances." *Journal of Public Economics* 88 (9-10): 1893–1915.

Bertrand, Marianne, Emir Kamenica, and Jessica Pan. 2015. "Gender Identity and Relative Income within Households." *Quarterly Journal of Economics* 130 (2): 571–614.

Bils, Mark, Peter J. Klenow, and Ciane Ruane. 2021. "Misallocation or Mismeasurement?" *Journal of Monetary Economics* 124: S39–S56.

Blundell, Richard, Pierre-Andre Chiappori, Thierry Magnac, and Costas Meghir. 2007. "Collective Labour Supply: Heterogeneity and Non-participation." *Review of Economic Studies* 74 (2): 417–45.
Bobonis, Gustavo J. 2009. "Is the Allocation of Resources within the Household Efficient? New Evidence from a Randomized Experiment." *Journal of Political Economy* 117 (3): 453–503.
Bourguignon, François, Martin Browning, Pierre-André Chiappori, and Valérie Lechene. 1993. "Intra Household Allocation of Consumption: A Model and Some Evidence from French Data." *Annales d'Economie et de Statistique*: 137–56.
Bronson, Mary Ann. 2014. "Degrees Are Forever: Marriage, Educational Investment, and Lifecycle Labor Decisions of Men and Women." Unpublished.
Browning, M., and P.-A. Chiappori. 1998. "Efficient Intra-household Allocations: A General Characterization and Empirical Tests." *Econometrica* 66 (6)*:* 1241–78.
Browning, Martin, Pierre-André Chiappori, and Valérie Lechene. 2010. "Distributional Effects in Household Models: Separate Spheres and Income Pooling." *Economic Journal* 120 (545): 786–99.
Bubb, Ryan, and Patrick L. Warren. 2020. "An Equilibrium Theory of Retirement Plan Design." *American Economic Journal: Economic Policy* 12 (2): 22–45.
Calvo, Paula A., Ilse Lindenlaub, and Ana Reynoso. 2024. "Marriage Market and Labour Market Sorting." *Review of Economic Studies* 91 (6): 3316–61.
Cesarini, David, Erik Lindqvist, Matthew J. Notowidigdo, and Robert Östling. 2017. "The Effect of Wealth on Individual and Household Labor Supply: Evidence from Swedish Lotteries." *American Economic Review* 107 (12): 3917–46.
Cherchye, Laurens, Bram De Rock, and Frederic Vermeulen. 2009. "Opening the Black Box of Intrahousehold Decision Making: Theory and Nonparametric Empirical Tests of General Collective Consumption Models." *Journal of Political Economy* 117 (6): 1074–1104.
Chetty, Raj, Nathaniel Hendren, Patrick Kline, and Emmanuel Saez. 2014. "Where Is the Land of Opportunity? The Geography of Intergenerational Mobility in the United States." *Quarterly Journal of Economics* 129 (4): 1553–1623.
Chiappori, Pierre-André. 1988. "Rational Household Labor Supply." *Econometrica* 56 (1): 63–90.
Chiappori, Pierre-André, Monica Costa Dias, and Costas Meghir. 2018. "The Marriage Market, Labor Supply, and Education Choice." *Journal of Political Economy* 126 (S1): S26–S72.
Chiappori, Pierre-André, Bernard Fortin, and Guy Lacroix. 2002. "Marriage Market, Divorce Legislation, and Household Labor Supply." *Journal of Political Economy* 110 (1): 37–72.
Chiappori, Pierre-Andre, and Maurizio Mazzocco. 2017. "Static and Intertemporal Household Decisions." *Journal of Economic Literature* 55 (3): 985–1045.
Chiappori, Pierre-André, and Costas Meghir. 2015. "Intrahousehold Inequality." In *Handbook of Income Distribution*, Vol. 2, edited by Anthony B. Atkinson and François Bourguignon, 1369–1418. Elsevier.
Choi, James J., David Laibson, and Brigitte C. Madrian. 2011. "$100 Bills on the Sidewalk: Suboptimal Investment in 401 (k) Plans." *Review of Economics and Statistics* 93 (3): 748–63.
Choukhmane, Taha. 2024. "Default Options and Retirement Saving Dynamics." Unpublished.
Choukhmane, Taha, Lucas Goodman, and Cormac O'Dea. 2025. *Data and Code for: "Efficiency in Household Decision-Making: Evidence from the Retirement Savings of US Couples."* Nashville, TN: American Economic Association; distributed by Inter-university Consortium for Political and Social Research, Ann Arbor, MI. https://doi.org/10.3886/E216286V1.
Conlon, John J., Malavika Mani, Gautam Rao, Matthew W. Ridley, and Frank Schilbach. 2021. "Learning in the Household." NBER Working Paper 28844.
Cosaert, Sam, Alexandros Theloudis, and Bertrand Verheyden. 2022. "Togetherness in the Household." *American Economic Journal: Microeconomics* 15 (1): 529–79.
Cronin, Julie-Anne, and Joseph Gray-Hancuch. 2024. "Barriers to Claiming Education Tax Credits for Low-Income Students." Office of Tax Analysis Working Paper 125.
Dauphin, Anyck, Abdel-Rahmen El Lahga, Bernard Fortin, and Guy Lacroix. 2011. "Are Children Decision-Makers within the Household?" *Economic Journal* 121 (553): 871–903.
Dauphin, Anyck, Bernard Fortin, and Guy Lacroix. 2018. "Is Consumption Efficiency within Households Falsifiable?" *Review of Economics of the Household* 16: 737–66.
Del Boca, Daniela, and Christopher Flinn. 2012. "Endogenous Household Interaction." *Journal of Econometrics* 166 (1): 49–65.
Derby, Elena, Kathleen Mackie, and Jacob Mortenson. 2023. "Worker and Spousal Responses to Automatic Enrollment." *Journal of Public Economics* 223: 104910.
Dobridge, Christine, Paul Landefeld, and Jacob A. Mortenson. 2021. "Corporate Taxes and the Earnings Distribution: Effects of the Domestic Production Activities Deduction." FEDS Working Paper.
Doepke, Matthias, and Fabian Kindermann. 2019. "Bargaining over Babies: Theory, Evidence, and Policy Implications." *American Economic Review* 109 (9): 3264–3306.

Gathergood, John, Neale Mahoney, Neil Stewart, and Jörg Weber. 2019. "How Do Individuals Repay Their Debt? The Balance-Matching Heuristic." *American Economic Review* 109 (3): 844–75.
Goldstein, Markus, and Christopher Udry. 2008. "The Profits of Power: Land Rights and Agricultural Investment in Ghana." *Journal of Political Economy* 116 (6): 981–1022.
Gollin, Douglas, and Christopher Udry. 2021. "Heterogeneity, Measurement Error, and Misallocation: Evidence from African Agriculture." *Journal of Political Economy* 129 (1): 1–80.
Goodman, Lucas, Anita Mukherjee, and Shanthi Ramnath. 2023. "Set It and Forget It? Financing Retirement in an Age of Defaults." *Journal of Financial Economics* 148 (1): 47–68.
Gross, David B., and Nicholas S. Souleles. 2002. "Do Liquidity Constraints and Interest Rates Matter for Consumer Behavior? Evidence from Credit Card Data." *Quarterly Journal of Economics* 117 (1): 149–85.
Gu, Ran, Cameron Peng, and Weilong Zhang. Forthcoming. "The Gender Gap in Household Bargaining Power: A Revealed-Preference Approach." *Review of Financial Studies*.
Hertzberg, Andrew. 2024. "Time-Consistent Individuals, Time-Inconsistent Households." *Journal of Finance* 79 (6): 3821–57.
Internal Revenue Service. 2024. *Population Files 1996–2022*. Department of Treasury, Washington, DC (accessed October 17, 2024).
Jordà, Òscar, Katharina Knoll, Dmitry Kuvshinov, Moritz Schularick, and Alan M. Taylor. 2019. "The Rate of Return on Everything, 1870–2015." *Quarterly Journal of Economics* 134 (3): 1225–98.
Kaiser Family Foundation. 2015. *Employer Health Benefits*. Kaiser Family Foundation.
LaFave, Daniel, and Duncan Thomas. 2017. "Extended Families and Child Well-Being." *Journal of Development Economics* 126: 52–65.
Ligon, Ethan. 2002. "Dynamic Bargaining in Households (with Application to Bangladesh)." Unpublished.
Lise, Jeremy, and Ken Yamada. 2019. "Household Sharing and Commitment: Evidence from Panel Data on Individual Expenditures and Time Use." *Review of Economic Studies* 86 (5): 2184–2219.
Low, Corinne. 2024. "The Human Capital-Reproductive Capital Trade-Off in Marriage Market Matching." *Journal of Political Economy* 132(2): 552–76. Unpublished.
Low, Hamish, Costas Meghir, Luigi Pistaferri, and Alessandra Voena. 2018. "Marriage, Labor Supply and the Dynamics of the Social Safety Net." NBER Working Paper 24356.
Lundberg, Shelly, and Robert A. Pollak. 1993. "Separate Spheres Bargaining and the Marriage Market." *Journal of Political Economy* 101 (6): 988–1010.
Lundberg, Shelly J., Robert A. Pollak, and Terence J. Wales. 1997. "Do Husbands and Wives Pool Their Resources? Evidence from the United Kingdom Child Benefit." *Journal of Human Resources* 32 (3): 463–80.
Lusardi, Annamaria. 2008. "Financial Literacy: An Essential Tool for Informed Consumer Choice?" NBER Working Paper 14084.
Lusardi, Annamaria, and Olivia S. Mitchell. 2011. "Financial Literacy and Planning: Implications for Retirement Well-Being." NBER Working Paper 17078.
Lusardi, Annamaria, and Olivia S. Mitchell. 2023. "The Importance of Financial Literacy: Opening a New Field." *Journal of Economic Perspectives* 37 (4): 137–54.
Madrian, Brigitte C., and Dennis F. Shea. 2001. "The Power of Suggestion: Inertia in 401(k) Participation and Savings Behavior." *Quarterly Journal of Economics* 116 (4): 1149–87.
Mazzocco, Maurizio. 2007. "Household Intertemporal Behaviour: A Collective Characterization and a Test of Commitment." *Review of Economic Studies* 74 (3): 857–95.
Rangel, Marcos, and Duncan Thomas. 2019. "Decision-Making in Complex Households." NBER Working Paper 26511.
Rauh, Joshua D., Irina Stefanescu, and Stephen P. Zeldes. 2020. "Cost Saving and the Freezing of Corporate Pension Plans." *Journal of Public Economics* 188: 104211.
Reynoso, Ana. 2024. "The Impact of Divorce Laws on the Equilibrium in the Marriage Market." *Journal of Political Economy* 132 (12): 4155–4204.
Roth, Jonathan, Pedro H. C. Sant'Anna, Alyssa Bilinski, and John Poe. 2023. "What's Trending in Difference-in-Differences? A Synthesis of the Recent Econometrics Literature." *Journal of Econometrics* 235 (2): 2218–44.
Saez, Emmanuel. 2010. "Do Taxpayers Bunch at Kink Points?" *American Economic Journal: Economic Policy* 2 (3): 180–212.
Schaner, Simone. 2015. "Do Opposites Detract? Intrahousehold Preference Heterogeneity and Inefficient Strategic Savings." *American Economic Journal: Applied Economics* 7 (2): 135–74.
Schultz, T. Paul. 1990. "Testing the Neoclassical Model of Family Labor Supply and Fertility." *Journal of Human Resources* 25 (4): 599–634.

Thomas, Duncan. 1990. "Intra-household Resource Allocation: An Inferential Approach." *Journal of Human Resources* 25 (4): 635–64.

Topoleski, John J., and Elizabeth A. Myers. 2020. *Worker Participation in Employer-Sponsored Pensions: Data in Brief*. Congressional Research Services.

Udry, Christopher. 1996. "Gender, Agricultural Production, and the Theory of the Household." *Journal of Political Economy* 104 (5): 1010–46.

US Department of Treasury. 2024. *Historical Tax Files* (accessed October 17, 2024).

Voena, Alessandra. 2015. "Yours, Mine, and Ours: Do Divorce Laws Affect the Intertemporal Behavior of Married Couples?" *American Economic Review* 105 (8): 2295–2332.

Walther, Selma. 2018. "Noncooperative Decision Making in the Household: Evidence from Malawi." *Journal of Development Economics* 134: 428–42.